

CLINICAL STUDY PROTOCOL

Protocol title	A multicentre, open-label, Phase 3, randomised controlled trial of duvelisib versus investigator's choice of gemcitabine or bendamustine in patients with relapsed/refractory nodal T cell lymphoma with T follicular helper (TFH) phenotype (TERZO)
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Study phase	3
EU CT number	2024-516605-23-00
Sponsor	Secura Bio Limited 32 Molesworth Street Dublin 2 Ireland
Supersedes	Version 1.0, dated 18 July 2024

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Sponsor Signatory:

I have read this protocol for Study SBI-0145-304, Version 2.0 and I approve the design of this study:

Signed by:
David Sidransky

Signer Name: David Sidransky
Signing Reason: I approve this document
Signing Time: 28-Oct-2024 | 05:55 PDT
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Investigator Agreement:

Secura Bio Limited.
Protocol No. SBI-0145-304, Version 2.0

I have read Protocol SBI-0145-304, a multicentre, open-label, Phase 3, randomised controlled trial of duvelisib versus investigator's choice of gemcitabine or bendamustine in patients with relapsed/refractory nodal T cell lymphoma with T follicular helper (TFH) phenotype (TERZO).

I will conduct the study as detailed herein and in compliance with International Council for Harmonisation Guidelines for Good Clinical Practice and applicable regulatory requirements and to inform all who assist me in the conduct of this study of their responsibilities and obligations.

Principal Investigator

Date

Printed Name:

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LIST OF ABBREVIATIONS

Abbreviation	Definition
¹⁸ FDG	[¹⁸ F] fluorodeoxyglucose
5PS	5-point scale
AE	Adverse event
AITL	Angioimmunoblastic T cell lymphoma
ALC	Absolute lymphocyte count
ALCL	Anaplastic large-cell lymphoma
ALT	Alanine aminotransferase
ANC	Absolute neutrophil count
AST	Aspartate aminotransferase
AUC	Area under the curve
B2M	β-2-microglobulin
β-hCG	β-human chorionic gonadotropin
BID	Twice daily
C	Cycle
C1D1	Cycle 1 Day 1
CEC	Central Ethics Committee
CLL	Chronic lymphocytic leukaemia
CMV	Cytomegalovirus
CR	Complete response
CRO	Contract Research Organisation
CRP	C-reactive protein
CRR	Complete response rate
CT	Computed tomography
CTCAE	Common Terminology Criteria for Adverse Events
CTCL	Cutaneous T cell lymphoma
CYP3A	Cytochrome P450 3A
D	Day
DMC	Data Monitoring Committee
DNA	Deoxyribonucleic acid
DOR	Duration of response
DRESS	Drug reaction with eosinophilia and systemic symptoms
EATL	Enteropathy-associated T cell lymphoma
ECG	Electrocardiogram
ECOG	Eastern Cooperative Oncology Group
eCRF	Electronic case report form
EORTC	European Organisation for Research and Treatment of Cancer
EOT	End of Treatment

Abbreviation	Definition
EU	European Union
FAS	Full analysis set
FDA	Food and Drug Administration
GCP	Good Clinical Practice
G-CSF	Granulocyte-colony stimulating factor
GI	Gastrointestinal
HBcAb	Hepatitis B core antibody
HBsAg	Hepatitis B surface antigen
HBV	Hepatitis B virus
HCVAb	Hepatitis C antibody
HIV	Human immunodeficiency virus
HR	Hazard ratio
HRQoL	Health-related quality of life
HSTCL	Hepatosplenic T cell lymphoma
HSV	Herpes simplex virus
HTLV-1	Human T-lymphotropic virus type 1
ICF	Informed consent form
ICH	International Council for Harmonisation
IEC	Independent Ethics Committee
Ig	Immunoglobulin
IHC	Immunohistochemistry
iNHL	Indolent non-Hodgkin lymphoma
IRB	Institutional Review Board
IRC	Independent Review Committee
IV	Intravenous(ly)
KM	Kaplan-Meier
LDH	Lactate dehydrogenase
LDi	Longest transverse diameter of a lesion
MedDRA	Medical Dictionary for Regulatory Activities
MRI	Magnetic resonance imaging
NCI	National Cancer Institute
NOS	Not otherwise specified
ORR	Objective response rate
OS	Overall survival
PD	Progressive disease
PET	Positron emission tomography
PFS	Progression-free survival
P-gp	P-glycoprotein

Abbreviation	Definition
PI3K	Phosphoinositide-3-kinase
PJP	<i>Pneumocystis jirovecii</i> pneumonia
PO	Oral(ly)
PPD	Cross product of the LD _i and perpendicular diameter
PR	Partial response
PTCL	Peripheral T cell lymphoma
QD	Once daily
QoL	Quality of life
QTcF	QT interval corrected with Fridericia's method
R/R	Relapsed/refractory
SAE	Serious adverse event
SCT	Stem cell transplantation
SJS	Stevens-Johnson syndrome
SD	Stable disease
SD _i	Shortest axis perpendicular to the LD _i
SmPC	Summary of Product Characteristics
SPD	Sum of the product of the perpendicular diameters for multiple lesions
SUV	Standardised uptake value
TEAE	Treatment-emergent adverse event
TEN	Toxic epidermal necrolysis
TESAE	Treatment-emergent serious adverse event
TFH	T follicular helper
UK	United Kingdom
ULN	Upper limit of normal
US	United States
VZV	Varicella zoster virus
WBC	White blood cell count
WOCP	Women/woman of childbearing potential
WHO-ATC	World Health Organization Anatomical, Therapeutic and Chemical

AMENDMENT 1 (16-OCT-2024)

Study SBI-0145-304 has not yet been initiated or submitted to any regulatory agencies. As the date of this amendment, no patients have been enrolled in the study.

The purpose of this amendment is to correct or clarify the following items based on feedback gathered from third-party vendors and potential investigators during protocol feasibility activities:

- In Section 1.3 Schedule of Activities, the following edits/clarifications were made:
 - Biomarker assessments: Added new row “Whole exome sequencing” to be performed at Screening to align with the exploratory endpoints. Updated footnote “below” to clarify that whole exome sequencing may be performed using slides submitted for confirmation of disease.
 - Biomarkers assessments: Clarified that Circulating DNA/plasma will also be performed at Cycle 1 Day 1.
 - Footnote ‘f’: Clarified that for patients who will be administered bendamustine, vital signs will also be taken on Day 2 of each cycle corresponding to the infusion schedule.
 - Footnote ‘i’: Clarified that results from HIV tests taken ≥ 2 years prior to C1D1 will require retesting. (This clarification is also added to Section 8.3.1.)
- Section 4.10 End of Study: Further defined end of study definition for a patient and for the study.
- Section 5.1 Inclusion Criteria #5: Clarified that baseline organ function should be measured in patients who have not had transfusion support within 30 days of C1D1. (This clarification is also added to Section 6.3 and 6.3.2.)
- Section 5.2 Exclusion Criteria #12: Clarified the definition of ‘chronic’ for immunosuppressants or systemic steroid use as continuous treatment for at least 4 weeks.
- Section 6.1.3: Removed item ‘3’ for alignment and consistency with approved study medicine label.
- Section 6.3: Clarified that use concomitant palliative radiotherapy is allowed in the study.
- Section 7.1: Updated the reasons for discontinuation/withdrawal reasons to align with other areas of the protocol and case report form entries.
- Section 8.1.4: Clarified timing and scope of relevant pathology reports used for diagnosis that may be included as part of the patient’s medical history.
- Section 8.9.9: Re-ordered the section paragraphs for additional clarity. Updated the time period for pregnancy reporting (from date of signed informed consent, not start of study medicine) to align with AE and SAE reporting timelines.
- Appendix 9: EORTC QLQ-NHL HG29 quality of life questionnaire: Correct the omission of 11 questions.

Changes to the protocol

All changes made in the body of the protocol were updated in the synopsis as appropriate. Minor editorial changes for accuracy, clarity, or formatting were also made throughout the protocol and are not included in the above description(s). All changes will be provided in a tracked changes version of the protocol for full transparency.

Review requirements by IRB/IEC and Health Authorities

A copy of this amended protocol will be sent to the Institutional Review Board (IRBs)/ Independent Ethics Committee (IECs) and Health Authorities. The changes described in this amended protocol are considered non-substantial.

Summary of previous amendment(s): N/A.

1 PROTOCOL SUMMARY

1.1 Synopsis

Study Title A multicentre open-label Phase 3, randomised controlled trial of duvelisib versus investigator's choice of gemcitabine or bendamustine in patients with relapsed/refractory nodal T cell lymphoma with T follicular helper (TFH) phenotype (TERZO)
Protocol Number: SBI-0145-304
Phase: 3
Investigational Agent(s) Arm 1 (Experimental arm): Duvelisib Arm 2 (Comparator arm), one of the following: Gemcitabine or bendamustine
Study Population This study will enrol a total of 124 randomised patients (62 per arm) with relapsed/refractory (R/R) nodal T cell lymphoma with T follicular helper (TFH) phenotype that has progressed on ≥ 1 prior lines of anti-cancer therapy.
Study Duration Patients in the duvelisib arm will receive treatment until progressive disease (PD), unacceptable toxicity, until withdrawal criteria are met, or for a maximum of 3 years, whichever occurs first. Patients in the comparator arm will receive treatment for up to 6 cycles (gemcitabine: 28-day cycles; bendamustine: 21-day cycles) or until PD, unacceptable toxicity, or until withdrawal criteria are met, whichever occurs first. All patients will be followed for survival until the end of the study. It is anticipated that the study will enrol over approximately 30 months; the study duration is expected to be 48 months. The duration of individual participation will be approximately 24 months.
Study Centres Enrolment is anticipated at approximately 45 sites in the European Union (EU) and the United Kingdom (UK).
Study Design This is a Phase 3, randomised, open-label study designed to evaluate the efficacy and safety of duvelisib (Arm 1) compared to gemcitabine or bendamustine (Arm 2) therapy in patients with R/R nodal T cell lymphoma with TFH phenotype. <u>Randomisation</u> Patients will be randomised 1:1 to one of 2 treatment arms: • Arm 1 (experimental arm): duvelisib • Arm 2 (comparator arm): Investigator's choice of gemcitabine or bendamustine

Stratification Factors

Participants will be stratified by angioimmunoblastic T cell lymphoma (AITL) score (0-2 vs 3-4) and number of prior lines of therapy for T cell lymphoma (1 vs >1).

Schedule of Administration

Arm 1

Cycles 1 and 2: Duvelisib 75 mg orally (PO) twice daily (BID) in 28-day cycles.

Cycles 3+: Duvelisib 25 mg PO BID in 28-day cycles.

Arm 2 (one of the following regimens after randomisation and by investigator's choice)

- Gemcitabine 1200 mg/m² IV over 30 minutes on Day (D)1, D8, and D15 of each 28-day cycle for up to 6 cycles
- Bendamustine 90-120 mg/m² IV over 30 minutes on D1 and D2 of each 21-day cycle for up to 6 cycles. The dose of 120 mg/m² is preferred, but lower doses within the range of 90-120 mg/m² are allowed at the investigator's discretion, as medically appropriate.

Patients will be monitored continuously for toxicity while on study treatment. Toxicity will be assessed using the National Cancer Institute (NCI) Common Terminology Criteria for Adverse Events (CTCAE) Version 5.0.

Study Objectives

Primary Objective

- To evaluate the Independent Review Committee (IRC)-determined progression-free survival (PFS) benefit of duvelisib monotherapy as compared to investigator's choice of gemcitabine or bendamustine

Key Secondary Objective

- To evaluate the overall survival (OS) benefit of duvelisib monotherapy as compared to gemcitabine or bendamustine

Secondary Objectives

- To evaluate additional efficacy parameters (Investigator-assessed PFS, objective response rate [ORR], complete response rate [CRR], duration of response [DOR], proportion of patients who proceed to stem cell transplantation [SCT], PFS in patients who proceed to SCT) achieved with duvelisib monotherapy as compared to gemcitabine or bendamustine
- To evaluate the safety of duvelisib monotherapy as compared to gemcitabine or bendamustine
- To evaluate the impact of duvelisib monotherapy on quality of life (QoL) as compared to gemcitabine or bendamustine

Exploratory Objective

- To evaluate the outcome of treatment with duvelisib compared to gemcitabine or bendamustine by biomarker studies

Study Endpoints

Primary Endpoint

- PFS, defined as the time from randomisation to documented PD, according to the Lugano 2014 criteria ([Cheson 2014](#)) as assessed by the IRC, or death due to any cause

Key Secondary Endpoint

- OS, defined as the time from randomisation to death due to any cause

Secondary Endpoints

- Investigator-assessed PFS according to the Lugano 2014 criteria
- ORR, with objective response defined as best response of complete response (CR) or partial response (PR) according to the Lugano 2014 criteria as assessed by the IRC
- CRR, with CR according to the Lugano 2014 criteria as assessed by the IRC
- DOR, defined as the time from the first documented response (CR or PR) to the first documentation of PD according to the Lugano 2014 criteria as assessed by the IRC or death due to any cause (for patients with CR or PR only)
- Proportion of patients who proceed to SCT
- Investigator-assessed PFS in patients who proceed to SCT
- Adverse events (AEs) and abnormal laboratory values
- QoL (EORTC QLQ-C30, EQ5D, QLQ-NHL-HG29)

Exploratory Endpoints

- Outcomes associated with exploratory biomarkers including but not limited to:
 - AITL vs non-AITL
 - Whole exome sequencing
 - Circulating deoxyribonucleic acid (DNA)
 - AITL score (including β -2-microglobulin [B2M], age, C-reactive protein [CRP], and Eastern Cooperative Oncology Group [ECOG] performance status)

Inclusion Criteria

Patients may be eligible for inclusion in the study if they meet all of the following criteria:

1. ≥ 18 years of age at the time of signing informed consent
2. Pathologically confirmed nodal T cell lymphoma with TFH phenotype according to the criteria of the World Health Organization classification ([Swerdlow 2017](#), [Alaggio 2022](#)) including any one of AITL, follicular T cell lymphoma, and other nodal peripheral T cell lymphoma (PTCL) with a TFH phenotype

Note: Local pathology report should be reviewed by the medical monitor prior to randomisation; Slides, including positive staining for at least 2 TFH-related antigens (CD279/PD1, CD10, BCL6, CXCL13, ICOS, SAP, and CCR5), should be submitted for

central pathology review; central pathology review is not required prior to initiation of treatment.

3. Relapsed or refractory to at least 1 prior systemic, cytotoxic therapy for T cell lymphoma
4. Measurable disease as defined by Lugano 2014 criteria ([Cheson 2014](#)) for T cell lymphoma
Note: Confirmation of measurable disease by central review or by medical monitor is required prior to randomisation.
5. Must have the following baseline organ function (without transfusion support within 30 days of C1D1) as defined by the following laboratory parameters:

Laboratory Parameter	Cut-off
Haematology	
Haemoglobin	≥ 8.0 g/dL (without transfusion support)
Platelet count	$\geq 50 \times 10^9/L$
Absolute neutrophil count (ANC)	$\geq 1.0 \times 10^9/L$ (without growth factor support for ≥ 7 days for filgrastim and ≥ 14 days for pegylated growth factors)
CD4 lymphocyte count	$\geq 50/mm^3$ ($0.05 \times 10^9/L$)
Renal	
Serum creatinine	$\leq 2.0 \times$ the upper limit of normal (ULN) or creatinine clearance ≥ 30 mL/min (estimated by Chronic Kidney Disease Epidemiology Collaboration [CKD-EPI] or Cockcroft-Gault equation or measured)
Liver	
Total bilirubin	$\leq 1.5 \times$ ULN (patients with documented Gilbert's syndrome may have a bilirubin $> 1.5 \times$ ULN but $\leq 3 \times$ ULN)
Alanine aminotransferase (ALT)	$\leq 2.5 \times$ ULN
Aspartate aminotransferase (AST)	$\leq 2.5 \times$ ULN

6. ECOG performance status ≤ 2
7. Recovery to Grade ≤ 1 or baseline for any non-haematologic toxicities due to prior treatments, except for peripheral neuropathy (Grade ≤ 2) or alopecia
8. Washout of at least 14 days or 5 half-lives, whichever is longer, from any prior T cell lymphoma-directed therapy (including approved and investigational therapies) to the first dose of study drug (14 days for radiation therapy)
9. For women of childbearing potential (WOCBP): negative serum β -human chorionic gonadotropin (β -hCG) pregnancy test within 3 days before first treatment
Note: WOCBP is defined as a sexually mature woman who has not undergone surgical sterilisation or who has not been naturally postmenopausal for ≥ 12 consecutive months for women > 55 years of age only.
10. Male and female patients of reproductive potential (i.e., not surgically sterile, or female patients who are not postmenopausal) must be willing to use a highly effective method of contraception for the duration of study treatment and for at least 3 months after the last dose of study drug.

11. Signed and dated institutional review board (IRB)/independent ethics committee (IEC)/central ethics committee (CEC)-approved informed consent form before any screening procedures are performed

Exclusion Criteria

Patients are to be excluded from the study if they meet any of the following criteria:

1. Cutaneous-only disease
2. Received prior allogeneic transplant any time in the past or received autologous transplant within 60 days prior to the first dose of study drug
3. Received prior treatment with a phosphoinositide-3-kinase (PI3K) inhibitor
4. Prior exposure to planned study treatment investigator's choice therapy (gemcitabine or bendamustine) within 60 days prior to the first dose of study drug
5. Major surgery within 4 weeks prior to the first dose of study drug and/or expected major surgical intervention while receiving study treatment
6. Any known history of or current central nervous system involvement by T cell lymphoma
7. Patients with a known history of human immunodeficiency virus (HIV) seropositivity or history of active or latent hepatitis B or C. Patients with cured hepatitis C and/or serology consistent with hepatitis B immunity are not excluded from the study.

Note: Patients with a positive hepatitis B surface antigen [HBsAg] or hepatitis C antibody [HCVAb] in the absence of documented cure will be excluded. Patients with a positive hepatitis B core antibody [HBcAb] must have negative hepatitis B virus [HBV] DNA to be eligible, must receive prophylaxis with entecavir [or equivalent] concomitant with duvelisib treatment, and must be periodically monitored for HBV reactivation by institutional guidelines.

8. Active cytomegalovirus (CMV) infection
9. History of tuberculosis treatment within 2 years prior to screening
10. History of chronic liver disease
Note: Patients with non-alcoholic fatty liver disease (NAFLD) can be included if they meet the inclusion criteria for bilirubin, ALT, and AST.
11. Active uncontrolled alcohol or drug abuse that may impair compliance with study protocol.
12. Ongoing treatment with chronic immunosuppressants (e.g., cyclosporine) or systemic steroids > 20 mg of prednisone (or equivalent) once daily. Note: chronic is defined as continuous treatment for at least 4 weeks.

13. Ongoing treatment for systemic bacterial, fungal, or viral infection at screening
Note: Patients on antimicrobial, antifungal, or antiviral prophylaxis are not specifically excluded if all other inclusion/exclusion criteria are met
14. Administration of a live viral vaccine within 6 weeks of the first dose of study drug
15. Concurrent administration of medications or foods (e.g., grapefruit) that are strong inhibitors or inducers of cytochrome P450 3A (CYP3A)
16. Unable to receive prophylactic treatment for *Pneumocystis* at screening
17. Prior surgery or condition with gastrointestinal dysfunction that may significantly affect drug absorption (e.g., gastric bypass surgery, gastrectomy)

18. Female patients who are pregnant or breastfeeding
19. History of other malignancy that could affect compliance with the protocol or interpretation of results:
 - Patients with a history of curatively treated basal or squamous cell carcinoma or melanoma of the skin or in situ carcinoma of the cervix at any time prior to the study are eligible.
 - Patients with any malignancy appropriately treated with curative intent and the malignancy has been in remission without treatment for ≥ 2 years prior to enrolment are eligible.
 - Patients with low-grade, early-stage prostate cancer (Gleason score 6 or below, Stage 1 or 2) with no requirement for therapy at any time prior to study are eligible.
20. History of stroke, unstable angina, and/or myocardial infarction within the last 6 months prior to screening, New York Heart Association Class 3 or 4 congestive heart failure, or an uncontrolled arrhythmia
21. Unstable or uncontrolled medical condition (e.g., unstable cardiac function, unstable pulmonary condition, uncontrolled diabetes), including severe psychiatric disorder or any important medical condition or abnormal laboratory finding that would, in the investigator's judgement, increase the risk to the patient associated with his or her participation in the study
22. Known hypersensitivity to any of the study drugs or drug product ingredients
23. Concurrent participation in another interventional clinical trial

Statistics

Sample size calculation

This study employs an event-driven, group sequential design. The planned number of PFS events at the final analysis for PFS is approximately 108. This will provide approximately 90% power to detect a difference in PFS if the true hazard ratio (HR) of Arm 1 and Arm 2 is 0.53 (5.3 months vs 2.8 months in median PFS, respectively), based on a 1-sided stratified log-rank test controlling the overall type 1 error rate at 0.025.

One interim analysis for efficacy and futility is planned, which will be conducted at approximately 50% of the planned number of PFS events (approximately 54 events). The boundaries for efficacy and futility at the interim analysis will be the Hwang-Shih-DeCani alpha and beta spending functions with spending parameters $\gamma = -4$ (approximate O'Brien-Fleming design) and $\gamma = -2$, respectively.

Assuming a 10% annual drop-out rate with drop-out time exponentially distributed, the required sample size is approximately 124 patients. With an accrual rate of 0.09 patients per site per month, accrual of patients is projected to last approximately 30 months, and the study duration is expected to be approximately 48 months.

Statistical Analyses

Two PFS analyses are planned for this study. The first analysis is an interim analysis when approximately 50% of events have occurred to assess efficacy and futility (non-binding). Final PFS analysis will be conducted when the planned number of events has been reached. Primary PFS analysis of the study will apply a 1-sided stratified log-rank test for comparison of Arm 1

(duvelisib) and Arm 2 (gemcitabine or bendamustine) stratified by AITL score (0-2 vs 3-4), and number of prior lines of therapy for T cell lymphoma (1 vs >1), with Hwang-Shih-DeCani alpha spending functions with parameter gamma = -4 and overall significance level 2.5%.

Key secondary OS analysis will be conducted after ~92 events. OS analysis will apply a 1-sided stratified log-rank test stratified by AITL score (0-2 vs 3-4) and number of prior lines of therapy for T cell lymphoma (1 vs >1) for comparison of Arm 1 (duvelisib) and Arm 2 (gemcitabine or bendamustine). To control the overall type 1 error of 0.025, OS will be tested through a fixed hierarchical sequence testing only if the primary endpoint PFS has been significant.

This design provides approximately 80% power to detect a difference in OS if the true HR for Arm 1 and Arm 2 is 0.55 (18 months vs 10 months in median OS, respectively), based on a 1-sided stratified log-rank test controlling the type 1 error rate at 0.025.

Data Monitoring Committee

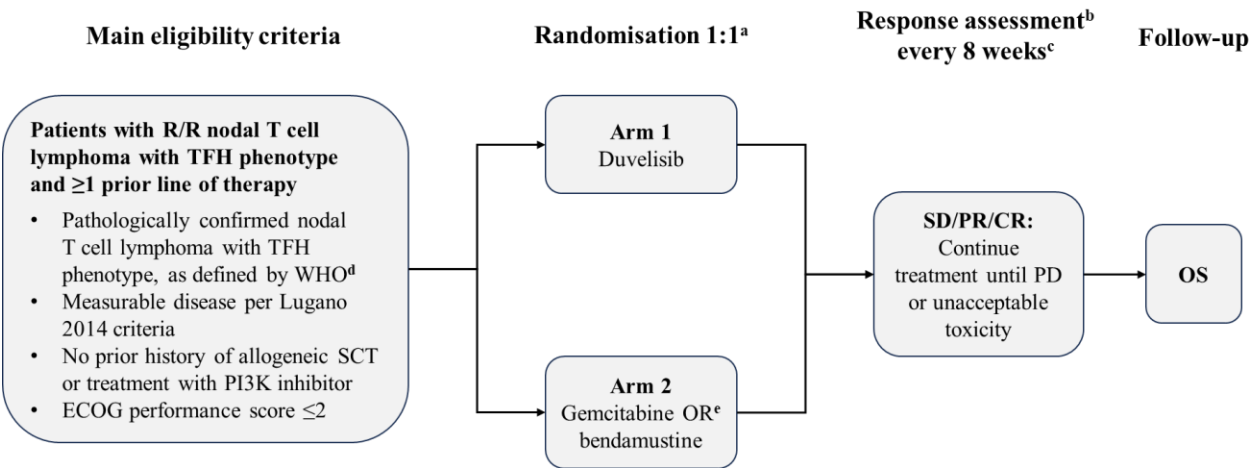
This study will utilise a data monitoring committee (DMC) for review of safety data and for the planned interim analysis of efficacy and futility. Composition and responsibilities of the DMC will be documented in a separate DMC Charter. The sponsor will make all final decisions regarding any changes in study conduct, considering the DMC's recommendations.

Independent Review Committee

Response will be assessed by an IRC through a central imaging vendor using the Lugano 2014 criteria ([Cheson 2014](#)). The IRC-assessed tumour response will be used for the primary and specified secondary endpoint analyses. The roles and responsibilities of the IRC are provided in a separate IRC charter.

1.2 Study Schema

Figure 1: Study Schema



Abbreviations: CR = complete response; ECOG = Eastern Cooperative Oncology Group; OS = overall survival; PD = progressive disease; PI3K = Phosphoinositide-3-kinase; PR = partial response; R/R = relapsed or refractory; SCT = stem cell transplantation; SD = stable disease; TFH = T follicular helper; WHO = World Health Organization

a Randomisation will be stratified by AITL score (0-2 vs 3-4) and number of prior lines of therapy for T cell lymphoma (1 vs >1).

b Response assessment per Lugano 2014 criteria ([Cheson 2014](#))

c In the bendamustine arm, response assessment occurs every 6 weeks until end of treatment.

d Definition by World Health Organization ([Swerdlow 2017](#), [Alaggio 2022](#))

e Investigator’s choice of comparator (gemcitabine or bendamustine) must be declared prior to randomisation.

1.3 Schedule of Activities

Table 1: Schedule of Activities

Activity	Screening	Cycle 1 (Duvelisib and gemcitabine: 1 Cycle = 28±3 days Bendamustine: 1 Cycle = 21±3 days)				Cycles 2-6 ^o	Cycles ≥7 only in Duvelisib arm (odd cycle numbers only): 1 Cycle = 28±3 days	EOT ^a	Safety F/U ^b	Durability of Response ⁱ	Survival F/U ^m
	D -30 to -1	D1 ^o	D8±1	D15±2	D22±3 ^o	D1 ^{o, q}	D1 ^q	≤ 7 days from DC	30 (+ 7) days from last dose	Q2M (±2 weeks) from EOT visit	Q3M (±2 weeks) from EOT visit
Screening assessments											
Informed consent ^c	X										
Inclusion/exclusion criteria	X										
Medical history/disease history/ prior therapy/ demographics	X										
TFH phenotype ^s	X										
AITL score	X										
ECOG performance status ^c	X	X				X	X				
Concomitant medications/procedures	X	X	X	X	X	X	X	X	X		
Randomisation and study drug administration											
Randomisation	D -7 to D 1										
Duvelisib administration (Arm 1)		75 mg BID PO for Cycles 1 and 2, followed by 25 mg BID PO continuously in 28-day cycles									

Activity	Screening	Cycle 1 (Duvelisib and gemcitabine: 1 Cycle = 28±3 days Bendamustine: 1 Cycle = 21±3 days)				Cycles 2-6 ^o	Cycles ≥7 only in Duvelisib arm (odd cycle numbers only): 1 Cycle = 28±3 days	EOT ^a	Safety F/U ^b	Durability of Response ⁱ	Survival F/U ^m
	D -30 to -1	D1 ^o	D8±1	D15±2	D22±3 ^o	D1 ^{o, q}	D1 ^q	≤ 7 days from DC	30 (+ 7) days from last dose	Q2M (±2 weeks) from EOT visit	Q3M (±2 weeks) from EOT visit
Gemcitabine ^k (Arm 2)		1200 mg/m ² IV on D1, D8 and D15 of each 28-day cycle, for up to 6 cycles									
Bendamustine ^k (Arm 2)		90-120 mg/m ² IV on D1 and D2 of each 21-day cycle, for up to 6 cycles									
Safety assessments											
AE assessment ^d	X										
Physical examination ^e	X	X ^f				X	X	X			
Vital signs ^g	X	X ^f	X	X	X	X	X	X			
ECG (12-lead) ^{c, h}	X										
PJP prophylaxis review ^p		X				X	X	X			
Laboratory assessments											
Blood chemistry ^{c, r}	X	X ^f	X	X	X	X	X	X			
CRP ^r	X	X ^f									
β2-microglobulin ^r	X	X ^f									
Haematology ^{c, r}	X	X ^f	X	X	X	X	X	X			
CD4 lymphocyte count	X										
Viral screening ⁱ	X										

Activity	Screening	Cycle 1 (Duvelisib and gemcitabine: 1 Cycle = 28±3 days Bendamustine: 1 Cycle = 21±3 days)				Cycles 2-6 ^o	Cycles ≥7 only in Duvelisib arm (odd cycle numbers only): 1 Cycle = 28±3 days	EOT ^a	Safety F/U ^b	Durability of Response ⁱ	Survival F/U ^m	
	D -30 to -1	D1 ^o	D8±1	D15±2	D22±3 ^o	D1 ^{o, q}	D1 ^q	≤ 7 days from DC	30 (+ 7) days from last dose	Q2M (±2 weeks) from EOT visit	Q3M (±2 weeks) from EOT visit	
CMV/EBV (via PCR)	As clinically indicated											
β-hCG pregnancy test ^j	X	X ^f	As clinically indicated									
Disease assessments												
Imaging and baseline/ response assessment ^l	X	Every 8 weeks until PD (for bendamustine arm: every 6 weeks)					Every 8 weeks until PD	X		Until PD or next therapy		
Bone marrow assessment ⁿ	X ⁿ	As clinically indicated										
Biomarker assessments												
Whole exome sequencing	X ⁿ											
Circulating DNA/plasma		C1D1 and every 8 weeks until PD (every 6 weeks until PD in the bendamustine arm)						X				
Quality of life assessments												
EORTC QLQ-C30 and QLQ-NHL-HG29		X				X	X	X				
EQ5D		X				X	X	X				
Survival follow-up												
Survival status ^m										X	X	

Abbreviations: ¹⁸FDG = [¹⁸F] fluorodeoxyglucose; AE = adverse event; AITL = Angioimmunoblastic T cell lymphoma; β-hCG = β-human chorionic gonadotropin; BID = twice daily; C = cycle; C1D1 = Cycle 1 Day 1; CMV = cytomegalovirus; CRP = C-reactive protein; CT = computed tomography; D = study day; DC = discontinuation; DNA = deoxyribonucleic acid; EBV = Epstein-Barr virus; ECG = electrocardiogram; ECOG = Eastern Cooperative Oncology Group; EOT = end of treatment; F/U = follow-up; HBcAb = Hepatitis B core antibody; HBsAg = Hepatitis B surface antigen; HCVAb = Hepatitis C antibody; HIV = human immunodeficiency virus; ICF = informed consent form; IV = intravenously; MRD = minimal residual disease; MRI = magnetic resonance imaging; PCR = polymerase chain reaction; PD = progressive disease; PET = positron emission tomography; PFS = progression-free survival; PJP = *Pneumocystis jirovecii* pneumonia; PO = orally; Q2M = every 2 months; Q3M = every 3 months; TFH = T follicular helper; WOCBP = women of childbearing potential

- a. An EOT visit should be performed within 7 days after the decision to discontinue study treatment, and in the comparator arm after completion of the full number of treatment cycles or the decision to discontinue study treatment. EOT assessments need not be repeated if performed within the previous 14 days (30 days for disease response assessments).
- b. A Safety F/U Visit will be performed 30 (+7) days from last dose of study drug. This visit should occur prior to the initiation of any subsequent anti-cancer therapy. This visit may be a telephone call if the patient does not require laboratory and/or other procedures related to any new or ongoing AEs, in which case a clinical visit will be required.
- c. Standard of care assessments that were performed prior to patient signing the ICF may be used at screening, provided they are performed within the screening window period.
- d. AEs should be collected from ICF signature through 30 (+7) days after last dose of study drug. Investigators are not obligated to actively seek information on AEs or SAEs after the Safety Follow-up visit (i.e. during the period of Durability of Response and Survival Follow-up). However, if the investigator learns of any SAE, including a death, at any time later than 30 (+7) days after last dose of study drug, and the investigator considers the event to be reasonably related to the study drug, the investigator must promptly notify the sponsor.
- e. A full physical examination at screening; all subsequent physical examinations are symptom directed and include assessment of disease-related constitutional symptoms.
- f. If performed within 3 days of C1D1, then screening results can be used for C1D1. For patients who will be administered bendamustine, vital signs will also be taken on Day 2 of each cycle corresponding to the infusion schedule.
- g. Vital signs measured in a semi-supine position after 5 minutes rest.
- h. A single ECG tracing is sufficient unless there is an abnormality, such as prolonged QTcF ≥ 480 msec, new arrhythmia, or other clinically significant finding. If an abnormality is observed, the ECG is to be repeated at the investigator's discretion. Additional on-treatment or follow-up ECGs should be performed at the investigator's discretion as clinically indicated.
- i. Including HIV (retesting required if HIV tests taken ≥ 2 years prior to C1D1)/ HTLV-1 (if no documented negative test in the past), HBsAg, HBcAb, and HCVAb, as well as CMV IgG and IgM. Perform CMV PCR, EBV PCR, and EBV serology as clinically indicated, per investigator discretion. Monitor patients with evidence of current or prior hepatitis infection for clinical or laboratory signs of hepatitis or reactivation.
- j. For WOCBP, a serum pregnancy test must be performed at screening and repeated pre-dose on C1D1; results must be available and confirmed to be negative before administration of the first study drug dose. If the screening pregnancy test is performed within 3 days before C1D1, it need not be repeated on C1D1. All other pregnancy tests should be performed as clinically indicated; if a positive urine test occurs, it must be confirmed by a serum test.
- k. Arm 2 (comparator arm): Investigator's choice of gemcitabine or bendamustine needs to be declared prior to randomisation
- l. Patients without PD at EOT will enter durability of response (including Survival F/U) following the Safety F/U Visit. Assessments will occur every 2 months after the EOT visit. For durability of response assessment, imaging scans of the neck, chest, abdomen, and pelvis are required for all patients. ¹⁸FDG-PET/CT is the preferred imaging modality. Where PET-CT is not feasible, CT with contrast or MRI may be used. Imaging assessments may be performed within 7

days before initiating the next cycle of therapy. Digital copies of all imaging/radiology must be sent to central imaging for independent review. Patients who discontinue treatment without PD should continue response assessments per the schedule above until confirmed disease progression, or start of a new anti-cancer therapy, whichever occurs first. After confirmed PD or start of a new anti-cancer therapy, patients will continue Survival F/U. Unscheduled imaging is permitted if clinically indicated per the discretion of the investigator.

- m. Patients with PD at EOT will have Survival F/U assessments that will occur every 3 months post-EOT visit or after confirmed PD or start of a new anti-cancer therapy, and until final analysis of PFS, 3 years after randomisation, or closure of the study by the sponsor. The Survival F/U assessment can be conducted by telephone interview. Information on initiation of other anti-cancer therapy (including start date, therapy type, and response on treatment) will also be collected. Patients may be followed for up to 3 years after randomisation.
- n. Bone marrow assessment is optional. Samples and corresponding pathology report should be sent for biomarker analysis if available. In addition, whole exome sequencing may be performed using slides submitted for confirmation of disease.
- o. Patients receiving bendamustine come in also on Day 2 of each treatment cycle to receive study treatment. Patients receiving bendamustine do not have a C1D22 visit. Patients receiving gemcitabine come in also on days 8 and 15 of each treatment cycle starting C2D1 to receive study treatment.
- p. Patients are required to receive *Pneumocystis* prophylaxis concomitant with duvelisib treatment. Selection of the *Pneumocystis* prophylaxis agent is per investigator discretion. Patients who are found to be intolerant of all available *Pneumocystis* prophylaxis may continue with study treatment at the discretion of the investigator. Patients should continue PJP prophylaxis following discontinuation of duvelisib until the CD4 lymphocyte count is $\geq 200/\text{mm}^3$.
- q. Treatment cycles will have a ± 3 days window compared to D1 of the previous cycle.
- r. Starting in C1D1, blood samples for evaluation of haematology and blood chemistry may be collected up to 3 days prior to the visit.
- s. For TFH designation, the neoplastic cells in histopathology specimen should express at least 2 TFH-related antigens, including CD279/PD1, CD10, BCL6, CXCL13, ICOS, SAP, and CCR5.

2 INTRODUCTION

2.1 Background

2.1.1 Peripheral T Cell Lymphoma

Peripheral T cell lymphomas (PTCL) are a group of rare, heterogeneous lymphoproliferative disorders that represent 10% to 15% of all non-Hodgkin's lymphomas ([d'Amore 2015](#)). There are currently multiple subtypes recognised by the World Health Organization ([Swerdlow 2017](#), [Alaggio 2022](#)). The primary nodal subtypes of PTCL, which include PTCL not otherwise specified (NOS), anaplastic large-cell lymphoma (ALCL), and angioimmunoblastic T cell lymphoma (AITL), together comprise > 80% of PTCL in Europe, while extranodal subtypes are primarily seen in Asia. Overall, the incidence of PTCL is higher in Asia than in Western countries ([Park 2014](#)). PTCL affects males slightly more than females, with a ratio of 1.9:1. The median age at diagnosis is 60 years ([Weiss 2023](#)).

AITL accounts for approximately 20% of all PTCL cases. This subtype generally occurs in older adults (at about 65 years of age) and has an aggressive course. The T follicular helper (TFH) cell has been identified as the cell of origin of AITL, and in the WHO classification AITL now represents a subgroup of PTCLs with a TFH phenotype. The cells express ≥ 2 TFH-associated markers including PD-1/CD279, CD10, BCL 6, CXCL13, ICOS, SAP and CCR5. Genetic studies have shown recurrent mutations that affect a significant proportion of cases of AITL. Importantly, many of the same genetic changes are observed in cases of PTCL-NOS that manifest a TFH phenotype. Nodal T cell lymphoma with TFH phenotype is an umbrella category created to highlight the spectrum of nodal lymphomas with a TFH phenotype including AITL, follicular T cell lymphoma, and other nodal PTCL with a TFH phenotype, with overlapping recurrent molecular/cytogenetic abnormalities recognised that potentially could impact therapy ([Swerdlow 2017](#), [Alaggio 2022](#)). Recently, the T cell project, an international prospective cohort study in patients with mature T cell and natural killer (NK) cell lymphomas, reported on a novel prognostic score for patients with AITL. The score comprises age ≥ 60 years, Eastern Cooperative Oncology Group (ECOG) performance status >2 , elevated $\beta 2$ -microglobulin (B2M), and elevated C-reactive protein (CRP). Among patients in the low (0-1), intermediate (2), and high (3-4) risk groups, the 5-year progression-free survival (PFS) estimates were 41%, 37%, and 13%, respectively. Among prognostic scores the AITL score had the strongest discriminant power ([Advani 2021](#)).

Patients with newly diagnosed PTCL are most commonly treated with anthracycline-based chemotherapy regimens such as cyclophosphamide, doxorubicin, vincristine, and prednisolone (CHOP) with or without the addition of etoposide. The exception is ALCL that is strongly CD30-positive; in this case, a regimen containing brentuximab vedotin in combination with cyclophosphamide, doxorubicin, and prednisolone (CHP) was shown to be superior ([Horwitz](#)

2022) and has received marketing authorisation in the EU and UK. These regimens are often followed by consolidation with high-dose chemotherapy and stem cell transplantation (SCT). However, the majority of patients develop relapsed or refractory (R/R) disease (Salah 2021; Zinzani 2016). In addition, co-existing co-morbidities often preclude SCT consolidation in older patients with PTCL.

Historically, patients with relapsed disease have had a dismal prognosis, with a median overall survival (OS) of less than 6 months. Per the European clinical practice guidelines, the only therapy with curative intent is SCT. Most patients are not eligible for transplant. It is recommended in the European and British Society of Haematology guidelines that patients with ALCL receive brentuximab vedotin and those with PTCL-NOS, AITL, enteropathy-associated T cell lymphoma (EATL) or hepatosplenic T cell lymphoma (HSTCL) receive other regimens, including single-agent bendamustine or gemcitabine. Other options for these patients include combination chemotherapy regimens like ICE (ifosfamide, cyclophosphamide and etoposide) (d'Amore 2015; NCCN guidelines V1.2023; Fox 2022).

Currently, no agent is approved in the EU or the UK for treatment of patients with R/R PTCL other than brentuximab vedotin for ALCL. In the United States (US), romidepsin, pralatrexate, and belinostat received accelerated approval by the Food and Drug Administration (FDA) for R/R PTCL. For romidepsin, the R/R PTCL indication was subsequently withdrawn from the US market when it failed to meet the primary endpoint of PFS in a confirmatory Phase 3 study in patients with first-line PTCL (BMS 2021). Sponsors of the remaining drugs have not initiated or reported the results of confirmatory follow-up randomised trials to gain full approval. None of these agents have gained approval in the EU or UK.

Bendamustine is a well-proven therapy for haematologic malignancies. It is a nitrogen mustard containing chloroethylamine, an alkylating group attached to a benzimidazole ring, a purine analogue, and butyric acid side chain that enables water solubility. Therefore, it only has partial cross-resistance to other alkylating agents. The BENTLY study evaluated bendamustine in patients with R/R peripheral and cutaneous T cell lymphoma and was conducted in France (Damaj 2013). In this study, patients received 120 mg/m² on Days 1 and 2 of a 21-day cycle for up to 6 cycles. The main histology in the study were AITL (53%) and PTCL-NOS (38%). Among this pretreated population, the study achieved an objective response rate (ORR) of 50%, complete response (CR) of 28%, partial response (PR) of 22%, as well as median duration of response (DOR), PFS, and OS of 3.5, 3.6 and 6.3 months, respectively. Although the study was not powered to show differences in response between the different histologic subtypes, the response rate and PFS between AITL and PTCL-NOS was not significantly different. AITL had an ORR of 69%, and PTCL-NOS had an ORR of 41% (Damaj 2013).

Gemcitabine is a pyrimidine antimetabolite with a low toxicity profile. It is another recommended therapy for R/R PTCL and was evaluated in a study in 44 patients. Among these patients, 30 had

cutaneous disease with mycosis fungoides, and 14 had PTCL-NOS. Responses included 12% CR and 59% PR ([Zinzani 2000](#)).

Bendamustine and Gemcitabine were part of the comparator arms in 2 recent studies in R/R PTCL, the ORACLE study (Efficacy and Safety of Oral Azacitidine Compared to Investigator's Choice Therapy in Patients With Relapsed or Refractory AITL; [Dupuis 2022](#)) and the LUMIERE study (Randomized Phase III Study of Alisertib or Investigator's Choice [Selected Single Agent] in Patients With Relapsed or Refractory PTCL; [O'Connor 2019](#)). In ORACLE, 86 patients with AITL or nodal follicular helper T cell lymphoma were randomized between treatment with CC-468 (n=42) to treatment with an investigator choice comparator of gemcitabine (n=24), bendamustine (n=16), or romidepsin (n=4). Patients received a median of 2 and 3.5 treatment cycles of gemcitabine and bendamustine, respectively. The ORR at 3 months, CR rate at 3 months, median PFS, and median OS were 43.2%, 22.7%, 2.8 months (95% confidence interval [CI] 1.87-4.83), and 10.3 months (95% CI 4.2-13.5), respectively. In LUMIERE, 30 patients with PTCL, including 8 with AITL and 15 with PTCL-NOS, received gemcitabine and 23 were evaluable for efficacy analysis. In this group of patients, the ORR, CR rate, and median PFS were 35%, 22%, and 57 days, respectively. Median OS was 367 days for all comparators combined (n=133, including 80 patients treated with pralatrexate, 23 with romidepsin, and 30 with gemcitabine). Median DOR with gemcitabine was 134 days.

To date, no agent has been shown to improve OS in patients with R/R PTCL. These treatments are associated with toxicities, and only one is globally available to date (brentuximab vedotin; [Laribi 2018](#)). Apart from a few patients who achieve CR, response durations remain brief. Therefore, PTCL remains a condition of high unmet medical need.

2.1.2 Duvelisib

Duvelisib inhibits PI3K- δ and PI3K- γ and reduces their cancer-promoting effects. PI3K is a lipid kinase involved in intracellular signal transduction in many cells. The catalytic subunits of PI3Ks exist as α , β , δ , and γ ; the δ and γ isoforms are preferentially expressed in leukocytes and are involved in innate and adaptive immune functions ([Clayton 2002](#); [Okkenhaug 2002](#); [Vanhaesebroeck 2010](#); [Fung-Leung 2011](#)). For patients with advanced haematologic malignancies, including T cell lymphomas, the pathways mediated by PI3K- δ,γ contribute to survival, proliferation, and differentiation of cancer cells. It is well recognised that cancer cells, through the expression of various cytokines, growth factors, and chemokines, recruit multiple cell types, including myeloid cells capable of differentiating into tumour-associated macrophages that promote angiogenesis and augment tumour growth ([Lewis 2006](#)). PI3K- δ and PI3K- γ also have roles in the establishment and maintenance of the tumour microenvironment that plays an important role in the development and maintenance of cancer cells ([Hanahan 2011](#)). Accordingly, PI3K- δ,γ inhibition would be expected to disrupt tumour maintenance and progression ([Schmid 2011](#)).

Duvelisib was approved in the US in September 2018 with a dosage of 25 mg orally BID and is indicated for the treatment of adult patients with relapsed or refractory chronic lymphocytic leukaemia (CLL) or small lymphocytic leukaemia after at least 2 prior therapies. Duvelisib was also approved in Europe and in UK in May 2021 for the treatment of R/R CLL after at least 2 prior therapies and for the treatment of follicular lymphoma that is refractory to at least 2 prior systemic therapies. In January 2023, the European Commission issued an orphan drug designation to duvelisib for the treatment of adult patients with PTCL. Duvelisib is a potent PI3K- δ,γ inhibitor being developed by Secura Bio as an orally administered therapeutic in haematologic malignancies, including PTCL. Duvelisib is mechanistically distinct from other agents either currently available or being developed for PTCL.

2.2 Study Rationale

The rationale for development of duvelisib as a cancer therapy is provided by the role of PI3K- δ and PI3K- γ activity in supporting the growth and survival of cancer cells and maintaining a tumour-supportive microenvironment. This rationale is further supported by data from clinical studies in which duvelisib has shown activity at doses ranging from 8 mg to 75 mg BID in various haematologic malignancies including CLL, indolent non-Hodgkin lymphoma (iNHL), cutaneous T cell lymphoma (CTCL), PTCL, and Richter's syndrome with an acceptable safety profile.

In a Phase 1 dose escalation study of duvelisib in patients with haematologic malignancies including PTCL, the maximum tolerated dose was identified as 75 mg twice daily (BID) ([Flinn 2018](#)). A cohort from this study was examined in a Phase 1, open-label study of duvelisib in patients with R/R PTCL or CTCL ([Horwitz 2018](#)). The Phase 2 PRIMO study (NCT03372057; sponsored by Secura Bio, Inc.) was an open-label, multicentre, parallel cohort study that evaluated duvelisib in patients with PTCL in a dose optimisation phase followed by a dose expansion phase ([Zinzani 2022](#); [Mehta-Shah 2023](#)). In the dose expansion phase of PRIMO, the duvelisib dose was 75 mg BID for 2 cycles (to maximise disease control), followed by 25 mg BID (to mitigate late toxicities) until progressive disease (PD) or unacceptable toxicity. Both the Phase 1 and Phase 2 data demonstrated promising efficacy of duvelisib in patients with R/R PTCL, with an ORR of 49% in the Phase 2 part of the study. Responders in the Phase 1 study had statistically significant decreases in the IL-12 inhibitory subunit β (p40) and the chemokine CXCL13, which is known to be produced by follicular helper-like PTCLs including AITL ([Horwitz 2018](#); [Mehta-Shah 2023](#)).

An interim analysis of the PRIMO study dose expansion phase (n=97, excluding 4 patients with rare histology; [Mehta-Shah 2023](#), data cutoff October 1, 2021) examined outcomes in patients with the 3 most common baseline histology subtypes (PTCL-NOS: N=52; AITL: N=30; ALCL: N=15; [Table 2](#)). In this analysis, the ORR (by Independent Review Committee [IRC] assessment) of duvelisib was higher in patients with AITL and PTCL-NOS (67% and 48%, respectively) than in patients with ALCL (13%). This corresponded to a longer median PFS in AITL and PTCL-NOS than in ALCL (9.1 and 3.5 months vs. 1.5 months). Although not powered for subset analyses, this

analysis suggests that the activity of duvelisib may not be uniform across different types of lymphomas. Therefore, this analysis provides the rationale for the inclusion of AITL and nodal T cell lymphoma with TFH phenotype in the patient population of the present Phase 3 study.

Table 2: Outcomes From the Phase 2 PRIMO Study Dose Expansion Phase, Stratified by Baseline Histology

Outcome (n=97)*	PTCL-NOS (N=52)	AITL (N=30)	ALCL (N=15)
ORR, n (%)	25 (48.1)	20 (66.7)	2 (13.3)
Best response, n (%)			
CR	14 (26.9)	16 (53.3)	2 (13.3)
PR	11 (21.2)	4 (13.3)	0 (NC)
Median PFS by IRC, months (95% CI)	3.5 (1.8, 8.1)	9.1 (6.2, NC)	1.5 (0.7, 1.7)
Median OS, months (95% CI)	10.9 (5.1, NC)	15.5 (9.5, 18.0)	4.8 (1.7, 15.7)
Median time to response (range)	1.7 (1.7, 0.5)	1.8 (1.9, 0.5)	2.6 (2.6, 1.3)
Median DOR by IRC, months (95% CI)	5.5 (2.0, 9.2)	8.8 (7.7, NC)	1.9 (1.9, 2.0)
Median DOR for patients achieving CR, months (95% CI)	7.4 (6.4, NC)	7.9 (3.3, NC)	1.9 (1.9, 2.0)

Abbreviations: AITL = angioimmunoblastic T cell lymphoma; ALCL = anaplastic large-cell lymphoma; CI = confidence interval; CR = complete response; DOR = duration of response; IRC = Independent Review Committee; NC = not calculated; ORR = objective response rate; OS = overall survival; PFS = progression-free survival; PR = partial response; PTCL-NOS = peripheral T cell lymphoma - not otherwise specified

* 4 patients in the dose expansion phase who had PTCL of minor histologies are not included in this table (1 patient each with enteropathy-associated T cell lymphoma, Epstein-Barr virus-associated lymphoma, diffuse large B-cell lymphoma, and T cell lymphoma). In the current analysis (n=97), 4 patients discontinued prior to first scheduled scan due to PD.

Source: [Mehta-Shah 2023](#) (data cutoff: October 1, 2021)

In the PRIMO study, a total of 156 patients with PTCL received treatment: 33 in the dose optimisation phase (20 patients [Cohort 1] received 25 mg BID; 13 patients [Cohort 2] received 75 mg BID) and 123 patients in the dose expansion phase. The following safety results are based on a data cut-off on 23 September 2022.

In the dose optimisation phase, all patients experienced at least 1 treatment-emergent adverse event (TEAE), and 24 patients (72.7%) experienced at least 1 treatment-emergent serious adverse event (TESAE). Twelve patients (36.4%) experienced a TEAE leading to discontinuation of duvelisib treatment, with 9 patients (27.3%) experiencing a treatment-related TEAE that led to discontinuation of duvelisib. Six patients (18.2%) experienced a TEAE with an outcome of death (all from Cohort 1).

In the dose expansion phase, 114 patients (92.7%) experienced at least 1 TEAE and 49 patients (39.8%) experienced at least 1 TESAE. Twenty-one patients (17.1%) experienced a TEAE leading to discontinuation of duvelisib, with 20 patients (16.3%) experiencing a treatment-related TEAE

that led to discontinuation of duvelisib. Sixteen patients (14.0%) experienced a TEAE with an outcome of death.

A summary of TEAEs in $\geq 20\%$ of subjects overall is presented in [Table 3](#).

Table 3: Most Frequent TEAEs ($\geq 20\%$ of Patients Overall) in the Phase 2 PRIMO Study

Preferred Term	Dose Optimisation		Expansion	All Patients (N=156) n (%)
	Cohort 1 25 mg BID (N=20) n (%)	Cohort 2 75 mg BID (N=13) n (%)	75/25 mg BID (N=123) n (%)	
Any TEAE	20 (100.0)	13 (100.0)	114 (92.7)	147 (94.2)
Rash	11 (55.0)	8 (61.5)	40 (32.5)	59 (37.8)
ALT increased	4 (20.0)	5 (38.5)	43 (35.0)	52 (33.0)
Diarrhoea	8 (40.0)	7 (53.8)	34 (27.6)	49 (31.4)
AST increased	3 (15.0)	4 (30.8)	40 (32.5)	47 (30.1)
Fatigue	7 (35.0)	5 (38.5)	35 (28.5)	47 (30.1)
Nausea	8 (40.0)	5 (38.5)	27 (22.0)	41 (26.3)
Pyrexia	6 (30.0)	8 (61.5)	24 (19.5)	39 (25.0)
Neutrophil count decreased	6 (30.0)	2 (15.4)	28 (22.8)	36 (23.1)
Anaemia	3 (15.0)	3 (23.1)	27 (22.0)	33 (21.2)

Abbreviations: ALT = alanine aminotransferase; AST = aspartate aminotransferase; BID = twice daily; PTCL = peripheral T cell lymphoma; TEAE = treatment-emergent adverse event

For additional data on safety, see Investigator's Brochure.

Data cutoff: 23 September 2022.

Based on the results of the Phase 2 PRIMO study, the current study is planned as a Phase 3 trial in patients with R/R nodal T cell lymphoma with TFH phenotype. Patients will receive treatment until PD, unacceptable toxicity, or other withdrawal criteria are met.

2.3 Benefit/Risk Assessment

2.3.1 Risk Assessment

Important identified risks of duvelisib include infections, diarrhoea/colitis, cutaneous reactions, pneumonitis, neutropenia, and alanine aminotransferase (ALT)/aspartate aminotransferase (AST) elevation. Potential risks include hepatotoxicity, cytochrome P450 3A (CYP3A) drug-drug interaction, and reproductive toxicity. These toxicities are managed using a combination of prophylaxis, supportive care, and dose modifications. A summary is provided in [Table 4](#) below; for additional information, refer to the Investigator's Brochure.

For information on the individual risks of the comparator treatments, refer to the Summary of Product Characteristics (SmPC) of bendamustine and gemcitabine.

Table 4: Risk Summary and Mitigation Strategy

Potential Risk of Clinical Significance*	Summary of Data/Rationale for Risk	Mitigation Strategy
Duvelisib Risks and Mitigations		
Infections	Occur commonly, particularly pneumonia and upper respiratory tract infections. Serious, including fatal (29/822; 3.5%), infections occurred in 26.4% (217/822) of patients with haematologic malignancies receiving duvelisib monotherapy. Serious, including fatal (3/763; 0.4%), PJP occurred in 1.2% (9/763) of patients taking duvelisib. CMV reactivation/infection occurred in 1.3% (10/763) of patients taking duvelisib.	- Follow the haematologic organ function regarding inclusion and exclusion criteria (Sections 5.1 and 5.2). - Regular physical examinations, vital signs and laboratory monitoring according to Section 8.2. - PJP prophylaxis and antiviral prophylaxis should be taken according to Section 6.3.1. - Management of duvelisib according to Section 6.2.
Diarrhoea/colitis	Serious, including fatal (1/822; 0.1%), diarrhoea or colitis occurred in 14.6% (120/822) of patients with haematologic malignancies receiving duvelisib monotherapy.	- Ensure enrolment of patient population per the inclusion and exclusion criteria (Section 8.2). - Monitoring for these events by regular visits, physical examinations, and safety laboratory monitoring according to Section 8.2. - Management of diarrhoea/colitis is outlined in Section 6.2.
Cutaneous reactions	Serious, including fatal (2/822; 0.2%), cutaneous reactions occurred in 4.4% (36/822) of patients with haematologic malignancies receiving duvelisib monotherapy. Fatal cases included DRESS and TEN.	- Patients with cutaneous-only lymphoma and patients known to have hypersensitivity to duvelisib are excluded. - Regular physical examinations and visits according to the Schedule of Activities (Table 1). - Management recommendations of cutaneous reactions according to Section 6.2.
Pneumonitis	Serious, including fatal (2/822; 0.2%), pneumonitis occurred in 2.9% (24/822) of patients with haematologic malignancies receiving duvelisib monotherapy.	- Per the exclusion criteria, patients with unstable or uncontrolled medical conditions, e.g., unstable pulmonary conditions are excluded. - Regular monitoring of vital signs and physical examinations will be performed according to the Schedule of Activities (Table 1). - Management of pneumonitis on study according to Section 6.2.
Neutropenia	Neutropenia $\geq$ Grade 3 occurred in 20.1% (165/822) of patients with haematologic malignancies receiving duvelisib monotherapy.	Patients being enrolled need to have an ANC of $\geq 1.0 \times 10^9/L$ without support for 7 days for filgrastim/14 days for pegylated growth

Potential Risk of Clinical Significance*	Summary of Data/Rationale for Risk	Mitigation Strategy
		factors to be included on the study (Section 5.1). - Regular blood counts are obtained to monitor for neutropenia per the Schedule of Activities (Table 1). - Growth factor support is permitted (Section 6.3.2). - Management guidelines for neutropenia on study according to Section 6.2.
ALT/AST elevation/ hepatotoxicity	TESAEs of ALT/AST elevation occurred in 1.5% (12/822, each) of patients with haematologic malignancies receiving duvelisib monotherapy. No findings of acute liver failure or death associated with liver failure have been observed in any clinical study with duvelisib. Transaminase elevations were generally asymptomatic and reversible upon stopping drug. Patients dosed with duvelisib in clinical studies have experienced elevated ALT in combination with elevated bilirubin, typically in the context of a complicated medical history. Elevations generally occurred early in duvelisib dosing, after which duvelisib was held and the events resolved.	- Patients need to have adequate liver function parameters per the inclusion criteria (Section 5.1), and patients with a known history of chronic liver disease are excluded (Section 5.2). - Regular physical examinations and liver-related laboratory tests are obtained during the study according to the Schedule of Activities (Table 1). - Management of abnormal liver laboratory parameters or liver toxicity according to Section 6.2.
CYP3A drug-drug interaction	CYP3A4 has been identified as the primary contributor to the metabolism of duvelisib. In clinical studies, exposure (AUC) to duvelisib increased approximately 295% in the presence of the strong CYP3A inhibitor ketoconazole, and decreased by 82% in the presence of rifampin, a CYP3A inducer. Systemic exposure to other medications that are substrates for CYP3A4 may be increased in patients receiving duvelisib. Co-administration with duvelisib increases the AUC of a sensitive CYP3A4 substrate which may increase the risk of toxicities of these drugs.	Concomitant use of drugs or foods that are strong inhibitors or inducers of CYP3A are not allowed during study treatment with duvelisib (Sections 5.2 and 6.4.4).
Reproductive toxicity	Based on findings in animals and its mechanism of action, duvelisib can cause foetal harm when administered to a pregnant woman.	WOCBP need a negative serum β-hCG pregnancy test before first treatment (Section 5.1).

Potential Risk of Clinical Significance*	Summary of Data/Rationale for Risk	Mitigation Strategy
	There are no available human data informing the associated risk of reproductive toxicity with duvelisib.	Male and female patients of reproductive potential must be willing to use a highly effective method of contraception for the duration of study treatment and for at least 3 months after the last dose of study drug (Section 5.1).
Comparator Risks and Mitigations		
Risks associated with gemcitabine	See gemcitabine SmPC.	- Similar to the mitigations listed for duvelisib, the study has specific inclusion and exclusion criteria as well as baseline requirements for organ function which enables safe administration of the comparator drugs. - There are frequent visits to evaluate the patients' vital signs, physical examinations, and laboratory testing to ensure mitigation of risks associated with the comparator drugs, see Schedule of Activities (Table 1). - Management of toxicities for these agents should be in accordance with the applicable SmPCs.
Risks associated with bendamustine	See bendamustine SmPC.	

Abbreviations: ALT = alanine aminotransferase; ANC = absolute neutrophil count; AST = aspartate aminotransferase; AUC = area under the curve; β -hCG = β -h = β -human chorionic gonadotropin; CI = confidence interval; CMV = cytomegalovirus; CYP3A = cytochrome P450 3A; DRESS = drug reaction with eosinophilia and systemic symptoms; HR = hazard ratio; NE = not estimable; OS = overall survival; PJP = *Pneumocystis jirovecii* pneumonia; SmPC = Summary of Product Characteristics; TEN = toxic epidermal necrolysis; TESAE = treatment-emergent serious adverse event; WOCBP = women of childbearing potential

* Based upon an updated long-term follow-up analysis report for study IPI-145-07, which compared duvelisib and ofatumumab in patients with chronic lymphocytic leukaemia or small lymphocytic lymphoma (CLL/SLL), we present the following information regarding OS, which was not mature in the data submitted with the original New Drug Application: In the updated data set, 90 (out of 159) patients from the ofatumumab arm crossed over to duvelisib, after disease progression. The median follow-up time for OS by the reverse Kaplan-Meier (KM) method was 63 months overall. OS was not statistically significantly different between the two treatment arms, although it appears longer for ofatumumab (HR 1.09; 95% CI, 0.79, 1.51; $p=0.59$). In the duvelisib arm the median OS was 52.3 months (95% CI, 41.8, 68.0), and in the ofatumumab arm was 63.3 months (95% CI, 41.2, NE). Although unfavourable to duvelisib, it should be noted that the CI of the HR includes the number 1, indicating there was no observed statistical difference in survival. The large confidence interval indicates a lack of precision of the estimated hazard ratio of 1.09 and the median survival estimates. With 5-years of follow-up, the mean survival times were 41.6 and 42.0 months for duvelisib and ofatumumab, respectively. For further information on side effects, refer to the duvelisib Summary of Product Characteristics.

2.3.2 Benefit Assessment

R/R PTCL has a dismal prognosis with an OS of usually less than 6 months. Therapeutic options for this population are rare and treatment success remains limited. Pralatrexate, a folate analogue, and belinostat and romidepsin, intravenously (IV) administered histone deacetylase inhibitors, have been approved in the US, but not in the EU or the UK, for the treatment of R/R PTCL based on ORR ranging from 25% to 38% in single-arm studies. Brentuximab vedotin, a CD30-directed

antibody-drug conjugate, also administered IV, is indicated for systemic ALCL, a subtype of PTCL, also on the basis of a single-arm study in patients with R/R PTCL. In contrast, duvelisib displayed an ORR of 49% in a cohort of 101 patients with R/R PTCL including 34% complete responses, which are much higher rates than previously observed in such a patient population.

A comparison of outcomes of single-arm studies with duvelisib, gemcitabine, and bendamustine as well as investigator choice arms in Phase 3 studies is presented in [Table 5](#).

2.3.3 Overall Benefit-Risk Conclusion

Clinical study results to date support a favourable benefit-risk ratio of duvelisib for patients with haematologic malignancies. The risks are consistent with the adverse event (AE) profile for the PI3K inhibitor drug class and AEs experienced by patients undergoing treatment for advanced cancers. Duvelisib monotherapy has demonstrated encouraging activity in R/R PTCL, and more pronounced activity in the AITL subgroup, seen in a Phase 2 PTCL study (PRIMO) and a Phase 1 study in haematologic malignancies. Consistent with the Phase 2 study, the duvelisib starting dose in this Phase 3 study will be 75 mg BID until the first response assessment (2 cycles) in an attempt to debulk the tumour more efficiently. Subsequently, the dose will be reduced to 25 mg BID to anticipate and reduce toxicities of later times of onset. This strategy is designed to reduce the risk of toxicity while maintaining clinical benefit.

Overall, the potential benefit to a patient with R/R nodal T cell lymphoma with TFH phenotype, including AITL, with few treatment options and historically a dismal prognosis outweighs any potential risks.

Table 5: Phase 2 Single-Arm Study and Phase 3 Investigator Choice Arm Results for Duvelisib, Gemcitabine, and Bendamustine in R/R PTCL

	PRIMO Duvelisib (N=82) ^a	PRIMO (PTCL-NOS only) Duvelisib N=52	PRIMO (AITL only) Duvelisib N=30	Gemcitabine (N=20) ^b	BENTLY Bendamustine (N=55) ^c	IC arm in Phase 3 studies	
						ORACLE (Azacitidine vs IC in R/R TFH PTCL) ^d	LUMIERE (Alisertib vs IC in R/R PTCL) ^e
N in IC arm	NA	NA	NA	NA	NA	44 (24 Gemcitabine 16 bendamustine 4 romidepsin)	30 Gemcitabine (23 evaluable for efficacy)
Disease type	PTCL-NOS: 63.4% AITL: 36.6%	PTCL-NOS: 100%	AITL: 100%	PTCL: 100%	PTCL-NOS: 38% AITL: 53%	AITL/TFH: 100% (90% confirmed)	PTCL-NOS: 50% AITL: 27% ALCL: 3%
Prior lines of therapy, median (range)	3 (1-9)	NA	NA	3 (2-5)	1 (1-3)	2 (IQR: 1-2)	2 (1-7)
Efficacy							
ORR, %	55	48	67	55	50	43%***	35
CR, %	37	27	53	30	28	23%***	22
PR, %	18	21	13	25	22	21%***	13
Median DOR, months	7.6	5.5	8.8	NA	3.5	NA**	4.4
Median PFS, months	4.37	3.5	9.1	NA	3.6	2.8	1.9
Median OS, months	11.4	10.9	15.5	–	6.2	10.3	12.0*

Abbreviations: AITL = angioimmunoblastic T cell lymphoma; ALCL = anaplastic large-cell lymphoma; BID = twice daily; CR = complete response; DOR = duration of response; IC = investigator choice; IQR = interquartile range; NA = Not available; ORR = objective response rate; OS = overall survival; PD = progressive disease; PFS = progression-free survival; PR = partial response; PTCL-NOS = peripheral T cell lymphoma - not otherwise specified; R/R = relapsed/refractory; SCT = stem cell transplantation; TFH = T follicular helper

a 75/25 mg BID expansion cohort; interim analysis (N=101), [Mehta-Shah 2023](#), minus 4 patients with minor PTCL histologies minus 15 patients with ALCL

b [Zinzani 2010](#); c [Damaj 2013](#); d [Dupuis 2022](#); e [O'Connor 2019](#)

* For the full investigator choice arm with 133 patients (30 gemcitabine, 80 pralatrexate, 23 romidepsin)

** Median treatment duration of 2 cycles (gemcitabine), 3.5 cycles (bendamustine), and 13.5 cycles (romidepsin) received

*** ORR, CR, and PR are at 3 months

3 STUDY OBJECTIVES, ENDPOINTS AND ESTIMANDS

3.1 Objectives and Endpoints

Objectives	Endpoints
Primary	
To evaluate the IRC-determined PFS benefit of duvelisib monotherapy as compared to investigator's choice of gemcitabine or bendamustine	PFS, defined as the time from randomisation to documented PD, according to the Lugano 2014 Criteria (Cheson 2014) as assessed by the IRC, or death due to any cause
Key Secondary	
To evaluate the OS benefit of duvelisib monotherapy as compared to gemcitabine or bendamustine	OS, defined as the time from randomisation to death due to any cause
Secondary	
To evaluate additional efficacy parameters (e.g. Investigator-assessed PFS, ORR, complete response rate [CRR], DOR, proportion of patients who proceed to SCT, PFS in patients who proceed to SCT) achieved with duvelisib monotherapy as compared to gemcitabine or bendamustine	- Investigator-assessed PFS according to the Lugano 2014 criteria - ORR, with objective response defined as best response of CR or PR according to the Lugano 2014 criteria as assessed by the IRC - CRR, with CR according to the Lugano 2014 criteria as assessed by the IRC - DOR, defined as the time from the first documented response (CR or PR) to the first documentation of PD according to the Lugano 2014 criteria as assessed by the IRC or death due to any cause (for patients with CR or PR only) - Proportion of patients who proceed to SCT - Investigator-assessed PFS in patients who proceed to SCT
To evaluate the safety of duvelisib monotherapy as compared to gemcitabine or bendamustine	AEs and abnormal laboratory values
To evaluate the impact of duvelisib monotherapy on QoL as compared to gemcitabine or bendamustine	QoL (EORTC QLQ-C30, EQ5D, QLQ-NHL-HG29)

<i>Exploratory</i>	
To evaluate the outcome of treatment with duvelisib compared to gemcitabine or bendamustine by biomarker studies	Outcomes associated with exploratory biomarkers including but not limited to: - AITL vs non-AITL - Whole exome sequencing - Circulating DNA - AITL score (including B2M, age, CRP, and ECOG performance status)

Abbreviations: AE = adverse event; AITL = angioimmunoblastic T cell lymphoma; B2M = β -2-microglobulin; CRP = C-reactive protein; CRR = complete response rate; DNA = deoxyribonucleic acid; DOR = duration of response; ECOG = Eastern Cooperative Oncology Group; EORTC = European Organisation for Research and Treatment of Cancer; IRC = independent review committee; PD = progressive disease; PFS = progression-free survival; ORR = objective response rate; OS = overall survival; SCT = stem cell transplantation

3.2 Estimands

3.2.1 Primary Estimand of Progression-Free Survival

Primary Trial Objective: To evaluate the IRC-determined PFS benefit of duvelisib monotherapy to investigator's choice of gemcitabine or bendamustine.

Estimand Scientific Question of Interest: What is the effect on PFS of assigning subjects to duvelisib vs. gemcitabine or bendamustine?

This primary estimand is the main clinical quantity of interest to be estimated in this study, which is defined by the following 5 attributes:

- Population: subjects with R/R nodal T cell lymphoma with TFH phenotype as defined by the inclusion and exclusion criteria
- Treatment: duvelisib monotherapy versus gemcitabine or bendamustine
- Variable (endpoint): PFS (PD assessed by the IRC)
- Population-level summary: Kaplan-Meier estimates of PFS, hazard ratio (HR) of duvelisib vs. gemcitabine or bendamustine
- Intercurrent events and handling strategies ([Table 6](#)): treatment discontinuation, use of subsequent anti-cancer therapy and stem cell transplantation

Table 6 Intercurrent Events and Handling Strategies – PFS

Intercurrent Events	Name of Strategy for Addressing Intercurrent Events and Its Intercurrent Events Description
Treatment discontinuation (due to AE or reasons other than AE or worsening of disease).	Treatment policy strategy: use time to PD or death, regardless of whether or not treatment discontinuation had occurred.
Use of subsequent anti-cancer therapy.	Hypothetical strategy: subjects are censored at the last disease assessment showing no evidence of PD before the use of subsequent anti-cancer therapy.
Stem cell transplantation	Hypothetical strategy: subjects are censored at the date of SCT

Abbreviations: AE = adverse event; PD = progressive disease; PFS = progression-free survival

3.2.2 Primary Estimand of Overall Survival

The primary estimand of OS is defined by the following 5 attributes:

- Population: subjects with R/R nodal T cell lymphoma with TFH phenotype
- Treatment: duvelisib vs. gemcitabine or bendamustine
- Variable (endpoint): OS
- Population-level summary: Kaplan-Meier estimates of OS, HR of duvelisib vs. gemcitabine or bendamustine
- Intercurrent events and handling strategies ([Table 7](#)): treatment discontinuation, use of subsequent anti-cancer therapy and stem cell transplantation, death due to competing risks

Table 7 Intercurrent Events and Handling Strategies – OS

Intercurrent Events	Name of Strategy for Addressing Intercurrent Events and Its Intercurrent Events Description
Treatment discontinuation (due to AE or reasons other than AE or worsening of disease)	Treatment policy strategy: use time to death, regardless of whether or not treatment discontinuation had occurred.
Use of subsequent anti-cancer therapy	Treatment policy strategy: use time to death, regardless of whether or not subsequent anti-cancer therapy has been used.
Death due to competing risks	Composite variable strategy: consider death due to reasons other than PD or treatment-emergent AEs as an OS event.
Stem cell transplantation	Hypothetical strategy: subjects are censored at the date of SCT

Abbreviations: AE = adverse event; OS = overall survival; PD = progressive disease

4 STUDY DESIGN

4.1 Overview of Study Design

This is a multicentre, open-label Phase 3, randomised controlled trial of duvelisib vs investigator's choice of gemcitabine or bendamustine in patients with R/R nodal T cell lymphoma with TFH phenotype. This study will be conducted at approximately 45 sites in the EU and the UK.

The study will be conducted in several phases: screening phase, treatment phase, safety follow-up phase, durability of response phase, and survival follow-up phase. The duration of individual participation will be approximately 24 months.

At screening, each participant will be randomised to receive either duvelisib or comparator (gemcitabine or bendamustine). The choice of the comparator will be declared by the site prior to randomisation.

Throughout the study, efficacy, safety, biomarkers, and QoL will be assessed at the timepoints indicated in the Schedule of Activities ([Table 1](#)). A study schema is available in [Figure 1](#).

The clinical database will be locked after all study data are reconciled (i.e., “cleaned”) for all patients who have participated in the study. In addition, the database will be frozen at the time of the interim analysis.

A Data Monitoring Committee (DMC) will be commissioned for review of safety data and for the interim analysis for efficacy and futility.

4.2 Screening Phase

All patients will be screened for study eligibility within 30 days before the first study drug dose (i.e., baseline, Cycle 1 Day 1 [C1D1]). Patient eligibility will be reviewed by the medical monitor prior to randomisation. Patients who are determined to be eligible will be randomised to duvelisib or comparator (gemcitabine or bendamustine).

At screening, the site will determine the presence of measurable lymph node disease. Disease will be measured using ¹⁸F-fluorodeoxyglucose (¹⁸FDG)-positron emission tomography (PET)-computed tomography (CT) scanning (CT images along with PET should be diagnostic with contrast wherever possible). Where PET-CT is not feasible, CT with contrast or magnetic resonance imaging (MRI) may be used. Subsequently, for fluorodeoxyglucose (FDG)-avid PTCL, PET-CT will be used to assess disease status and for FDG-non-avid PTCL, CT with contrast or MRI will be appropriate substitutes. For each patient, only one of these modalities should be used to evaluate response, and the modality chosen for each individual patient should be the same throughout the duration of the study. For the purposes of patient management during the conduct of the study, the investigator's assessment of disease response will be used. Images will be submitted to an IRC for evaluation of efficacy endpoints.

At screening, AITL score ([Appendix 1](#)) is calculated.

4.2.1 Randomisation

Patients will be randomly assigned at a 1:1 ratio to one of 2 study arms as follows:

- Arm 1: Duvelisib 75 mg orally (PO) BID in 28-day cycles for Cycles 1 and 2, followed by duvelisib 25 mg PO BID in 28-day cycles for Cycles 3+
- Arm 2: One of the following regimens after randomisation and by investigator's choice (the regimen to be used after randomisation must be selected by the investigator prior to randomisation):
 - Gemcitabine 1200 mg/m² IV over 30 minutes on Day (D) 1, D8, and D15 of each 28-day cycle, for up to 6 cycles
 - Bendamustine 90-120 mg/m² IV over 30 minutes on D1 and D2 of each 21-day cycle, for up to 6 cycles. The dose of 120 mg/m² is preferred, but lower doses within the range of 90-120 mg/m² are allowed at the investigator's discretion, as medically appropriate.

4.2.2 Stratification

Randomisation will be stratified by AITL score (0-2 vs 3-4) and number of prior lines of therapy for T cell lymphoma (1 vs >1).

4.3 Treatment Phase

Response will be assessed every 8 weeks (every 6 weeks in the bendamustine arm until end of treatment, and then every 8 weeks) until PD.

Arm 1 (Duvelisib)

Patients will continue duvelisib treatment until PD, unacceptable toxicity, or other withdrawal criteria are met, or for a maximum of 3 years, whichever occurs first.

Dose modification criteria for duvelisib will be applied if the starting dose or a subsequent dose of duvelisib is deemed intolerable (see [Section 6.2](#)).

Patients with CR, PR, or SD will have their duvelisib dose maintained at the current dose level.

Patients with PD, patients who are not tolerating therapy, or patients who are otherwise not suitable to continue receiving therapy will discontinue duvelisib.

Arm 2 (Gemcitabine or Bendamustine)

Patients in Arm 2 will receive either gemcitabine or bendamustine according to the defined regimen for up to 6 cycles or until PD, unacceptable toxicity, or other withdrawal criteria are met, whichever occurs first.

4.4 Completion of Treatment

Patients will be considered to have completed study treatment if they receive the study drug until the full number of treatment cycles (comparator arm only) or the maximum of 3 years (duvelisib arm only), PD, or discontinuation for unacceptable toxicity, withdrawal of consent, or death. Patients will attend an end of treatment (EOT) visit within 7 days after receiving the last dose of study drug and will continue to be followed for other follow-up assessments specified in the Schedule of Activities ([Table 1](#)).

4.5 Safety Follow-up Phase

After discontinuation of study drug, all patients are to attend an EOT visit within 7 days after the decision to discontinue study treatment and then a Safety Follow-up visit 30 days after the last study drug dose. EOT assessments need not be repeated if performed within the previous 14 days (30 days for disease response assessments). Patients who have not experienced PD will continue to be followed for disease response as per [Table 1](#) until development of PD, as determined by the investigator.

4.6 Durability of Response and Survival Follow-up Phases

Patients who have not experienced PD at EOT will continue to be followed for disease response according to the schedule in [Table 1](#). Patients with documented PD, as determined by the investigator, will be followed on an every-3-month basis for survival until the final analysis of PFS, 3 years after randomisation, or closure of the study by the sponsor. Follow-up for survival may be conducted via telephone. Patients may be followed for up to 3 years after randomisation.

4.7 Rationale for Study Design

To reduce bias, and evaluate the clinical benefit of duvelisib, the study is a randomised controlled trial. The choice of comparator is consistent with the standard of care used in Europe and UK as discussed in [Section 2.1.1](#).

4.7.1 Patient Population Rationale

The study will include patients with R/R nodal T cell lymphoma with TFH phenotype per the analysis of the PRIMO study (see [Section 2.2](#) for details).

4.7.2 Dose Justification

In the Phase 1, dose escalation study in patients with haematologic malignancies including PTCL (IPI-145-02), duvelisib was administered BID at escalating doses of 8, 15, 25, 35, 50, 60, 75, and 100 mg, with the maximum tolerated dose identified as 75 mg BID. All patients with PTCL in this Phase 1 study received dose levels ≥ 60 mg; therefore, the activity of duvelisib in patients with PTCL at lower dose levels had not been evaluated. Accordingly, the dose optimisation phase of

the Phase 2 PTCL study (PRIMO; VS-0145-225) employed 2 dosing regimens that evaluated efficacy, safety, tolerability, and pharmacokinetics (PK) to select the recommended dose for the expansion phase of the study.

During the dose optimisation phase, patients in Cohort 1 received a starting dose of 25 mg BID with potential for escalation to 50 mg BID and then to 75 mg BID (on a per-patient basis, depending on the patient's response to, and tolerance of, therapy), and patients in Cohort 2 received 75 mg BID.

Data from the dose optimisation phase suggest that, while both starting doses are active, it appears that more patients discontinued prematurely in Cohort 1 due to more rapid disease progression and toxicity. This observation suggests that the higher starting dosage of 75 mg BID may result in a more rapid debulking of tumour and more rapid disease control. Therefore, to achieve both a rapid remission and ameliorate longer-term toxicities, the dosage for the dose expansion phase was set to 75 mg BID until the first response assessment (2 cycles), at which time the dose is reduced to 25 mg BID. As it is now known that the 25 mg BID dosage can be active in PTCL and later-onset toxicities may be prevented, a proactive decrease in the dose is justified.

The available results from the expansion phase of the Phase 2 PRIMO study suggest that this dosing strategy – i.e. duvelisib starting dose of 75 mg BID until the first response assessment (2 cycles) followed by 25 mg BID - is efficacious and has an acceptable safety profile. Therefore, this dosing strategy was also selected for the present Phase 3 study.

4.7.3 Blinding

Since duvelisib is an oral drug administered BID until disease progression while the comparator drugs are administered IV on certain days within a cycle and with a fixed duration of treatment, it is not possible to conduct a double-blinded study. To reduce bias in evaluation of responses, an IRC will be used. The IRC will be blinded to the treatment allocation.

4.7.4 Independent Review Committee

An IRC will be established to provide a blinded review of imaging data in order to provide expert interpretation of treatment response. The IRC will be composed of independent clinicians not otherwise involved in the study. The specifics of the IRC composition and processes will be described in a separate IRC charter.

4.7.5 Data Monitoring Committee

This study will utilise a DMC for review of safety data and for the planned interim analysis of efficacy and futility. Composition and responsibilities of the DMC will be documented in a separate DMC Charter. The sponsor will make all final decisions regarding any changes in study conduct, considering the DMC's recommendations.

4.8 Number of Patients

Approximately 124 patients with R/R nodal T cell lymphoma with TFH phenotype will be randomised in the study, with approximately 62 randomised patients each in Arms 1 and 2. For each patient who discontinues before receiving at least 1 cycle of study drug, an additional patient may be recruited and randomised.

4.9 Duration of Patient Participation

The duration of patient participation is dependent on the patient's tolerance of and response to study drug; it is anticipated that patients will participate in the study for about 24 months.

4.10 End of Study Definition

The end of the study for a given patient is defined as the date when the patient discontinues from follow-up assessments (OS).

The end of study overall is defined as 3 years after the date of the last follow-up visit of the last patient who entered the follow-up period, or at Sponsor's decision to end the study, whichever occurs first.

5 STUDY POPULATION

5.1 Inclusion Criteria

Patients eligible for inclusion in this study must fulfil all of the following criteria:

1. ≥ 18 years of age at the time of signing informed consent
2. Pathologically confirmed nodal T cell lymphoma with TFH phenotype according to the criteria of the World Health Organization classification ([Swerdlow 2017](#), [Alaggio 2022](#)) including any one of AITL, follicular T cell lymphoma, and other nodal peripheral T cell lymphoma (PTCL) with a TFH phenotype

Note: Local pathology report should be reviewed by the medical monitor prior to randomisation; Slides, including positive staining for at least 2 TFH-related antigens (CD279/PD1, CD10, BCL6, CXCL13, ICOS, SAP, and CCR5), should be submitted for central pathology review; central pathology review is not required prior to initiation of treatment.

3. Relapsed or refractory to at least 1 prior systemic, cytotoxic therapy for T cell lymphoma
4. Measurable disease as defined by Lugano 2014 criteria ([Cheson 2014](#)) for T cell lymphoma

Note: confirmation of measurable disease by central review or by medical monitor is required prior to randomisation.

5. Must have the following baseline organ function (without transfusion support within 30 days of C1D1) as defined by the following laboratory parameters:

Laboratory Parameter	Cut-off
Haematology	
Haemoglobin	≥ 8.0 g/dL (without transfusion support)
Platelet count	$\geq 50 \times 10^9/L$
Absolute neutrophil count (ANC)	$\geq 1.0 \times 10^9/L$ (without growth factor support for ≥ 7 days for filgrastim and ≥ 14 days for pegylated growth factors)
CD4 lymphocyte count	$\geq 50/mm^3$ ($0.05 \times 10^9/L$)
Renal	
Serum creatinine	$\leq 2.0 \times$ the upper limit of normal (ULN) or creatinine clearance ≥ 30 mL/min (estimated by Chronic Kidney Disease Epidemiology Collaboration [CKD-EPI] or Cockcroft-Gault equation or measured)
Liver	
Total bilirubin	$\leq 1.5 \times$ ULN (patients with documented Gilbert's syndrome may have a bilirubin $> 1.5 \times$ ULN but $\leq 3 \times$ ULN)
ALT	$\leq 2.5 \times$ ULN
AST	$\leq 2.5 \times$ ULN

6. ECOG performance status ≤ 2
7. Recovery to Grade ≤ 1 or baseline for any non-haematologic toxicities due to prior treatments, except for peripheral neuropathy (Grade ≤ 2) or alopecia
8. Washout of at least 14 days or 5 half-lives, whichever is longer, from any prior T cell lymphoma-directed therapy (including approved and investigational therapies) to the first dose of study drug (14 days for radiation therapy)
9. For women of childbearing potential (WOCBP): negative serum β -human chorionic gonadotropin (β -hCG) pregnancy test within 3 days before first treatment
- Note: WOCBP defined as a sexually mature woman who has not undergone surgical sterilisation or who has not been naturally postmenopausal for ≥ 12 consecutive months for women > 55 years of age only.
10. Male and female patients of reproductive potential (i.e., not surgically sterile, or female patients who are not postmenopausal) must be willing to use a highly effective method of contraception for the duration of study treatment and for at least 3 months after the last dose of study drug.
11. Signed and dated institutional review board (IRB)/independent ethics committee (IEC)/central ethics committee (CEC)-approved informed consent form before any screening procedures are performed

5.2 Exclusion Criteria

Patients are to be excluded from the study if they meet any of the following criteria:

1. Cutaneous-only disease
2. Received prior allogeneic transplant any time in the past or received autologous transplant within 60 days prior to the first dose of study drug
3. Received prior treatment with a phosphoinositide-3-kinase (PI3K) inhibitor
4. Prior exposure to planned study treatment investigator's choice therapy (gemcitabine or bendamustine) within 60 days prior to the first dose of study drug
5. Major surgery within 4 weeks prior to the first dose of study drug and/or expected major surgical intervention while receiving study treatment
6. Any known history of or current central nervous system involvement by T cell lymphoma
7. Patients with a known history of human immunodeficiency virus (HIV) seropositivity or history of active or latent hepatitis B or C. Patients with cured hepatitis C and/or serology consistent with hepatitis B immunity are not excluded from the study.

Note: Patients with a positive hepatitis B surface antigen [HBsAg] or hepatitis C antibody [HCVAb] in the absence of documented cure will be excluded. Patients with a positive hepatitis B core antibody [HBcAb] must have negative hepatitis B virus [HBV] DNA to be eligible, must receive prophylaxis with entecavir [or equivalent] concomitant with duvelisib treatment, and must be periodically monitored for HBV reactivation by institutional guidelines.

8. Active cytomegalovirus (CMV) infection
9. History of tuberculosis treatment within 2 years prior to screening
10. History of chronic liver disease

Note: Patients with non-alcoholic fatty liver disease (NAFLD) can be included if they meet the inclusion criteria for bilirubin, ALT, and AST.

11. Active uncontrolled alcohol or drug abuse that may impair compliance with study protocol.
12. Ongoing treatment with chronic immunosuppressants (e.g., cyclosporine) or systemic steroids > 20 mg of prednisone (or equivalent) once daily. Note: chronic is defined as continuous treatment for at least 4 weeks.
13. Ongoing treatment for systemic bacterial, fungal, or viral infection at screening
Note: Patients on antimicrobial, antifungal, or antiviral prophylaxis are not specifically excluded if all other inclusion/exclusion criteria are met
14. Administration of a live viral vaccine within 6 weeks of the first dose of study drug
15. Concurrent administration of medications or foods (e.g., grapefruit) that are strong inhibitors or inducers of CYP3A
16. Unable to receive prophylactic treatment for *Pneumocystis* at screening
17. Prior surgery or condition with gastrointestinal dysfunction that may significantly affect drug absorption (e.g., gastric bypass surgery, gastrectomy)
18. Female patients who are pregnant or breastfeeding

19. History of other malignancy that could affect compliance with the protocol or interpretation of results:
 - Patients with a history of curatively treated basal or squamous cell carcinoma or melanoma of the skin or in situ carcinoma of the cervix at any time prior to the study are eligible.
 - Patients with any malignancy appropriately treated with curative intent and the malignancy has been in remission without treatment for ≥ 2 years prior to enrolment are eligible.
 - Patients with low-grade, early-stage prostate cancer (Gleason score 6 or below, Stage 1 or 2) with no requirement for therapy at any time prior to study are eligible.
20. History of stroke, unstable angina, and/or myocardial infarction, within the last 6 months prior to screening, New York Heart Association Class 3 or 4 congestive heart failure, or an uncontrolled arrhythmia
21. Unstable or uncontrolled medical condition (e.g., unstable cardiac function, unstable pulmonary condition, uncontrolled diabetes), including severe psychiatric disorder or any important medical condition or abnormal laboratory finding that would, in the investigator's judgement, increase the risk to the patient associated with his or her participation in the study
22. Known hypersensitivity to any of the study drugs or drug product ingredients
23. Concurrent participation in another interventional clinical trial

5.3 Screen Failures

A screen failure occurs when a patient who has consented to participate in the study is not subsequently randomised. A minimal set of screen failure information is required to ensure transparent reporting and to respond to queries from regulatory authorities. Minimal information includes demography, screen failure details and reasons, eligibility criteria, and any serious adverse event (SAE).

In the event a screening assessment does not meet eligibility criteria but may be repeated in the screening window and it is considered by the investigator that the test results are not reliable, and that re-test is likely to meet eligibility criteria, the test may be repeated without need for re-screening the patient.

In the event a participant is a screen failure, the participant may be re-screened once.

6 STUDY DRUG REGIMEN

6.1 Dosing and Administration of Study Drug

All eligibility criteria must be met and documented prior to randomisation and prior to study drug administration. The study drug will be administered only to eligible patients under the supervision of the investigator or designated sub-investigator(s).

Arm 1

Duvelisib is to be self-administered by patients as oral capsule(s) supplied by the sponsor. Patients will receive duvelisib PO BID continuously in 28-day cycles.

On C1D1, the morning dose will be administered in the clinic. Patients should be instructed that each dose is to be taken at approximately the same time of day. Doses should be taken every 12 hours within ± 4 hours of the scheduled dose. Missed or vomited doses should not be taken or repeated.

Duvelisib should be swallowed whole with a glass of water (approximately 8 ounces or 240 mL). Duvelisib may be administered without regard to meals.

Dose reductions and interruptions for individual patients because of toxicity may be made based on the clinical judgement of the investigator (see Section 6.2).

Refer to the Pharmacy Manual for additional instructions regarding study drug administration.

Arm 2

The comparators (gemcitabine or bendamustine) will be administered IV in the clinic, according to the instructions in the respective SmPC.

Both bendamustine and gemcitabine are authorized drugs and included in European and British clinical treatment guidelines as single-agent treatment options for PTCL (d'Amore 2015, Fox 2022). The permissible range of 90-120 mg/m² for bendamustine is based on consensus panel recommendations published by Cheson 2016 and clinical experience (e.g., published by Reboursiere 2016 and Reboursiere 2018).

6.1.1 Dosing

Arm 1

Cycles 1 and 2: Duvelisib 75 mg PO BID in 28-day cycles

Cycles 3+: Duvelisib 25 mg PO BID in 28-day cycles

Arm 2

Investigator's choice of one of the following regimens:

- Gemcitabine: 1200 mg/m² IV over 30 minutes on D1, D8, and D15 of each 28-day cycle for up to 6 cycles

OR

- Bendamustine: 90-120 mg/m² IV over 30 minutes on D1 and D2 of each 21-day cycle for up to 6 cycles. The dose of 120 mg/m² is preferred, but lower doses within the range of 90-120 mg/m² are allowed at the investigator's discretion, as medically appropriate.

6.1.2 Handling of Missed Dosing

If a patient missed dosing in part of a study treatment cycle that was already initiated for events related or not related to the study drug (e.g. a required medical procedure or unanticipated personal emergency), the days missed will not be replaced. However, if a patient in the comparator arm would miss doses at the beginning of a cycle, (e.g. Days 1-7) because of a not study-related activity (e.g., vacation), then the next dose would be Day 1 of the next cycle (creating a delay between cycles). A maximum of 2 weeks of delay is allowable.

6.1.3 Study Drug Description

Duvelisib drug substance is a white to off-white crystalline powder. The duvelisib drug product is formulated in 2 different capsule strengths (25 mg and 15 mg; percent active drug in capsules is identical) for oral delivery with excipients (diluent, glidant, disintegrant, and lubricant) that are listed in the US FDA Inactive Ingredients Database for approved drug products and/or generally recognised as safe.

All investigational products and study materials will be manufactured, handled, stored, and provided to site personnel in accordance with current good manufacturing practice regulations and guidance, and International Council on Harmonization (ICH) guidelines for Good Clinical Practice (GCP).

Minimally, all investigational products or their immediate package will bear a label with the following information:

1. The name and place of business of the manufacturer, packer, or distributor (in accordance with 21 CFR 801.1).
2. The name of the investigational product.
3. Applicable statements per Annex VI of the REGULATION (EU) No 536/2014 OF THE EUROPEAN PARLIAMENT AND OF THE COUNCIL of 16 April 2014 on clinical trials on medicinal products for human use, and repealing Directive 2001/20/EC.

The investigational product label and/ or other labelling will describe all relevant contraindications, hazards, AEs, interfering substances or devices, warnings, and precautions.

Additional information is provided in the Pharmacy Manual.

For description of the comparator drug substances and drug products, refer to the respective SmPC.

6.1.4 Dispensing and Storage Instructions

Dispensing and storage instructions for duvelisib will be provided in the Pharmacy Manual.

Upon receipt at the study site, duvelisib should remain in the blister cards provided until use or dispensation. The blister cards should be stored at the study site at room temperature (15°C to 30°C or 59°F to 86°F). All excursions should be brought to Secura Bio's attention for assessment and authorisation for continued use. Expired drug is not to be dispensed.

For storage instructions for the comparators, refer to the respective SmPC.

6.1.5 Drug Accountability

Accountability for study drug at the site is the responsibility of the investigator. The investigator or delegated study staff will verify the integrity of the clinical study supplies (storage conditions, correct amount received, condition of shipment, kit numbers, etc.).

Records of duvelisib (used, lost, destroyed, and returned blister cards) must be maintained. Drug accountability records will indicate the drug's delivery to and receipt by the site, inventory at the site, dispensing to patients, use by each patient, and amount returned and/or disposed. Drug accountability records will be checked routinely by the study monitor.

Unused duvelisib will be destroyed locally per site standard operating procedures. If the site is unable to destroy locally, unused duvelisib will be returned to a location designated by Secura Bio. Unused duvelisib cannot be returned without approval and documentation by the clinical research associate. Destruction of duvelisib should not occur until the end of the study, unless otherwise authorised by Secura Bio.

The comparator drugs will be accounted for and disposed of according to the regulations and procedures at each local dispensing pharmacy. For details refer to the Pharmacy Manual.

6.2 Toxicity Management

6.2.1 Toxicity Grading Criteria

Toxicity grading will be assessed by the investigator using the National Cancer Institute (NCI) Common Terminology Criteria for Adverse Events (CTCAE) version 5.0 (see web page at <http://ctep.cancer.gov> for details).

6.2.2 Dose Modifications and Discontinuation

In the following, dose modification of duvelisib is described. For dose modification of the comparator drugs, refer to the respective SmPC.

If a patient has an AE considered related to duvelisib as defined in Section 8.9.5, then dose interruptions with possible modifications and toxicity management may be implemented as noted in Table 8. Adjustments to these guidelines may occur based on the clinical judgement of the investigator with notification to the medical monitor/sponsor. Additional information on the management of toxicities can be found in the Summary of Data and Guidance to the Investigator section of the Investigator's Brochure.

Table 8: Duvelisib Dose Modification and Toxicity Management

Treatment-Related Toxicity ^{a,b}	Adverse Reaction Grade	Duvelisib Dose Modification and Toxicity Management
Non-Haematologic Adverse Reactions		
Infections	Grade 3 or higher infection	- Withhold until resolved - Resume at the same or reduced dose (see Table 9)
	Clinical CMV infection or viremia (positive PCR or antigen test)	- Withhold until resolved - Resume at the same or reduced dose (see Table 9) - If duvelisib is resumed, monitor patients for CMV reactivation (by PCR or antigen test) at least monthly
	PJP	- For suspected PJP, withhold duvelisib until evaluated - For confirmed PJP, discontinue duvelisib
Non-infectious diarrhoea or colitis	Mild/moderate diarrhoea (Grade 1-2, up to 6 stools per day over baseline) and responsive to antidiarrheal agents OR Asymptomatic (Grade 1) colitis	- No change in dose - Initiate supportive therapy with antidiarrhoeal agents as appropriate - Monitor at least weekly until resolved
	Mild/moderate diarrhoea (Grade 1-2, up to 6 stools per day over baseline) and unresponsive to antidiarrheal agents	- Withhold until resolved - Initiate supportive therapy with enteric acting steroids (e.g., budesonide) - Monitor at least weekly until resolved - Resume at a reduced dose (see Table 9)
	Abdominal pain, stool with mucus or blood, change in bowel habits, peritoneal signs OR Severe diarrhoea (Grade 3, > 6 stools per day over baseline)	- Withhold until resolved - Initiate supportive therapy with enteric acting steroids (e.g., budesonide) or systemic steroids - Monitor at least weekly until resolved - Resume at a reduced dose (see Table 9) - For recurrent Grade 3 diarrhoea or recurrent colitis of any grade, discontinue
	Life-threatening	Discontinue

Treatment-Related Toxicity ^{a,b}	Adverse Reaction Grade	Duvelisib Dose Modification and Toxicity Management
Cutaneous reactions	Grade 1-2	- No change in dose - Initiate supportive care with emollients, antihistamines (for pruritus), or topical steroids - Monitor closely
	Grade 3	- Withhold until resolved - Initiate supportive care with emollients, antihistamines (for pruritus), or topical steroids - Monitor at least weekly until resolved - Resume at reduced dose (see Table 9) - If severe cutaneous reaction does not improve, worsens, or recurs, discontinue
	Life-threatening	Discontinue
	SJS, TEN, DRESS (any grade)	Discontinue
Pneumonitis without suspected infectious cause	Moderate (Grade 2) symptomatic pneumonitis	- Withhold - Treat with systemic steroid therapy - If pneumonitis recovers to Grade 0 or 1, duvelisib may be resumed at reduced dose (see Table 9) - If non-infectious pneumonitis recurs or patient does not respond to steroid therapy, discontinue
	Severe (Grade 3) or life-threatening pneumonitis	- Discontinue - Treat with systemic steroid therapy
ALT/AST elevation	3 to 5 × ULN (Grade 2)	- Maintain duvelisib dose - Monitor at least weekly until return to < 3 × ULN
	> 5 to 20 × ULN (Grade 3)	- Withhold duvelisib and monitor at least weekly until return to < 3 × ULN - Resume duvelisib at same dose (first occurrence) or at a reduced dose for subsequent occurrence (see Table 9)
	> 20 × ULN (Grade 4)	Discontinue duvelisib
Haematologic Adverse Reactions		
Febrile neutropenia	Grade 3-4	- Withhold duvelisib until afebrile and resolution of Grade 3 or Grade 4 neutropenia to Grade ≤ 2 (ANC > 1.0 × 10⁹/L) - Monitor ANC at least weekly until > 1.0 × 10⁹/L - Resume at same dose (first occurrence) or at a reduced dose for subsequent occurrence
Neutropenia	ANC 0.5 to 1.0 Gi/L	- Maintain duvelisib dose - Monitor ANC at least weekly

Treatment-Related Toxicity ^{a,b}	Adverse Reaction Grade	Duvelisib Dose Modification and Toxicity Management
	ANC less than 0.5 Gi/L	- Withhold duvelisib - Monitor ANC until $> 0.5 \times 10^9/L$ - Resume at same dose (first occurrence) or at a reduced dose for subsequent occurrence (see Table 9)
Thrombocytopenia	Platelet count 25 to < 50 Gi/L (Grade 3) with Grade 1 bleeding	- No change in dose - Monitor platelet counts at least weekly
	Platelet count 25 to < 50 Gi/L (Grade 3) with Grade 2 bleeding or Platelet count < 25 Gi/L (Grade 4)	- Withhold duvelisib - Monitor platelet counts until $\geq 25 \times 10^9/L$ and resolution of bleeding (if applicable) - Resume at same dose (first occurrence) or resume at a reduced dose for subsequent occurrence (see Table 9)

Abbreviations: ALT = alanine aminotransferase; ANC = absolute neutrophil count; AST = aspartate aminotransferase; CMV = cytomegalovirus; DRESS = drug reaction with eosinophilia and systemic systems; Gi = 1×10^9 ; PCR = polymerase chain reaction; PJP = *Pneumocystis jirovecii*; pneumonia; SJS = Stevens-Johnson syndrome; TEN = toxic epidermal necrolysis; ULN = upper limit of normal

a Treatment-related toxicity: related to study treatment as defined in Section 8.9.5

b Toxicity grades are defined per CTCAE version 5.0. Note if parameter is not defined by CTCAE, then AE grading criteria (Section [8.9.4](#)) should be utilised.

Duvelisib may be withheld up to 21 days for toxicity. Doses withheld for > 21 days due to treatment-related toxicity will result in permanent discontinuation from duvelisib unless the sponsor's medical monitor approves resuming treatment after investigator confirmation of acceptable benefit-risk assessment. Any patient who requires a dose level reduction to below dose level -3 during Cycles 1 and 2 and to below dose level -1 during Cycles 3+ (see [Table 9](#)) due to treatment-related toxicities will be permanently discontinued from duvelisib treatment.

Table 9: Stepwise Duvelisib Dose Reduction Levels for Patients Receiving 75 mg or 25 mg BID

Dose Level	Duvelisib (mg)
Starting Dose (Cycles 1 and 2)	75 BID
-1	50 BID
-2	25 BID
-3	15 BID
Starting Dose (Cycles 3+)*	25 BID
-1	15 BID

Abbreviations: BID = twice daily

* If the last administered dose in Cycle 2 was 15 mg BID, the starting dose in Cycle 3 is 15 mg BID.

Stepwise dose re-escalation (to 75 mg in Cycles 1 and 2 and 25 mg in Cycles 3+) may be considered per investigator discretion. No other dose levels are allowed.

In the event that duvelisib is discontinued due to any reason other than PD (e.g., AE), patients should continue disease response assessments until documentation of PD (as determined by the investigator), death, or new anti-cancer treatment.

6.3 Permitted Concomitant Medications and Procedures

Patients should receive full supportive care (including transfusion of blood and blood products and antibiotics, etc.) according to local standards and guidelines. Treatment for nausea, vomiting, and diarrhoea is permitted according to institutional guidelines. Concomitant medications, including concomitant palliative radiotherapy, received from 30 days before the screening visit through the Safety Follow-up visit will be recorded in the electronic case report form (eCRF). However, transfusion support is not permitted within 30 days of C1D1.

6.3.1 Antimicrobial Prophylaxis

Based on the duvelisib clinical experience to date, the following measures are recommended for patients receiving duvelisib:

- Patients are required to receive *Pneumocystis* prophylaxis concomitant with duvelisib treatment. Selection of the *Pneumocystis* prophylaxis agent is per investigator discretion. Patients who are found to be intolerant of all available *Pneumocystis* prophylaxis may continue with study treatment at the discretion of the investigator. Patients should continue PJP prophylaxis following discontinuation of duvelisib until the CD4 lymphocyte count is $\geq 200/\text{mm}^3$.
- Herpes simplex virus (HSV) and varicella zoster virus (VZV) infections have been observed with duvelisib; therefore, herpes (HSV/VZV) prophylaxis concomitant with treatment is recommended, per Investigator discretion according to institutional guidelines.
- Patients with a history of CMV infection or reactivation that required treatment, or history of other opportunistic infections should be monitored for reactivation per investigator standard practice. Prophylactic treatment per institutional guidelines is recommended for patients considered by the investigator to be at high risk for CMV reactivation.
- Antimicrobial prophylaxis is recommended for patients with a history of or considered at high risk for opportunistic infections and during periods of severe neutropenia. Choice of antimicrobial agent (antifungal, antibiotic, antiviral) is per the investigator's discretion.
- Investigators may consider administration of pneumococcal pneumonia vaccine prior to C1D1.

For patients in the comparator arm, antimicrobial prophylaxis should be as per institutional standard.

6.3.2 Transfusion and Growth Factor Support (Prophylaxis or Supportive Care)

During any time on treatment, blood cell transfusion (packed red blood cells or platelets) to maintain a patient's haemoglobin ≥ 7.0 mg/dL or platelets $\geq 10,000$ per μL is recommended. Transfusions may be used at any time, as clinically indicated, except within 30 days of C1D1.

Prophylactic use of growth factors such as granulocyte-colony stimulating factor (G-CSF) or pegylated G-CSF may be implemented if clinically indicated, in accordance with institutional guidelines and/or according to the National Comprehensive Cancer Network or American Society of Clinical Oncology (ASCO) practice guidelines for myeloid growth factors). Patients on a stable dose of erythropoietin to treat baseline anaemia may continue on this therapy at this dose.

6.3.3 Contraception and Pregnancy

At screening, male and female patients of reproductive potential (i.e., not surgically sterile or female patients who are not postmenopausal) must be willing to use a highly effective method of contraception for the duration of study treatment and for at least 3 months after the last dose of duvelisib. See [Appendix 2](#) for Contraceptive Guidance including examples of highly effective contraceptive methods. Pregnancy testing will be performed as shown in the Schedule of Activities ([Table 1](#)).

6.4 Excluded Concomitant Medications and Procedures

6.4.1 Vaccines

For all patients receiving duvelisib, the use of live or live attenuated vaccines is prohibited within 6 weeks prior to C1D1, during the treatment with duvelisib, and for 3 months after the last dose of duvelisib. For information on vaccination restrictions applicable to the comparator treatments, refer to the SmPCs of bendamustine and gemcitabine.

Investigators may consider administration of pneumococcal pneumonia vaccine prior to C1D1.

The use of inactivated (or killed) vaccines is allowed during the study, however patients and their physicians should be aware that the effectiveness of any vaccine administered concomitantly with duvelisib may be diminished. The ability to generate an immune response to any vaccine following the administration of duvelisib has not been studied.

6.4.2 Immunosuppressants

Patients are not to receive ongoing treatment with chronic immunosuppressants (e.g., cyclosporine) or systemic steroids for > 1 week at doses $>$ the equivalent of 20 mg prednisone QD, unless being used to treat treatment-emergent toxicity.

NOTE: Acute treatment for underlying autoimmune disorders (e.g., reactive airway disease, rheumatoid arthritis) with corticosteroid doses > 20 mg prednisone or equivalent QD for ≤ 1 week

is permitted during the study. Corticosteroid doses of ≤ 20 mg prednisone or equivalent QD are permitted during the study for physiological replacement or chronic treatment for underlying autoimmune disorders (e.g., reactive airway disease, rheumatoid arthritis).

6.4.3 Other Anti-Cancer Therapy or Investigational Agents

During study treatment period, additional anti-cancer therapy or other investigational agents are prohibited.

6.4.4 Medications or Food that Inhibit or Induce CYP3A (for Patients Receiving Duvelisib Only)

In vitro data indicate that duvelisib is mostly metabolised by CYP3A4. Therefore, concomitant use of drugs or foods that are strong inhibitors or inducers of CYP3A are not allowed during study treatment with duvelisib.

[Appendix 3](#) provides a list of medications known to be strong inhibitors or inducers of CYP3A. Please note that this appendix is not an exhaustive list of all medications that may modulate CYP3A activity.

Patients should avoid eating grapefruits, grapefruit-containing products, or Seville oranges. In addition, patients should avoid herbal supplements including, but not limited to, St. John's Wort throughout the study as this is known to be a strong inducer of CYP3A.

The sponsor should be contacted with any questions regarding concomitant use of medications that are thought to modulate CYP3A activity. The concomitant use of moderate or weak inhibitors may be allowed in selected circumstances.

6.5 Concomitant Medications to be Used with Caution

6.5.1 Medications that are Substrates of CYP3A or CYP2C8 (for Patients Receiving Duvelisib Only)

In vitro studies have demonstrated that duvelisib is an inhibitor of CYP3A4. Co-administration of duvelisib with midazolam, a sensitive CYP3A substrate, resulted in an approximate 4-fold increase in the midazolam AUC. Systemic exposure to other medications that are substrates for CYP3A may be increased in patients receiving duvelisib. Caution should be used if duvelisib is administered concomitantly with drugs that are substrates for CYP3A, particularly those with a narrow therapeutic range. Drugs that are substrates for CYP3A should be used only if medically necessary and therapeutic alternatives are not available.

Duvelisib is an inhibitor of CYP2C8 in vitro. Physiologically based pharmacokinetic modelling indicates the inhibitory effect of duvelisib on CYP2C8 substrates is not expected to be clinically meaningful at a duvelisib dose of 25 mg BID. The predicted mean AUC ratio was 1.02 for rosiglitazone (a CYP2C8 substrate) with and without duvelisib at 25 mg BID. Inhibition of

CYP2C8 metabolism appears to be negligible, and only minor changes in the systemic exposure to CYP2C8 substrates are anticipated in the presence of duvelisib. Medications that are metabolised via CYP2C8 may be used as medically indicated but with caution.

[Appendix 4](#) provides a list of medications known to be substrates of CYP3A or CYP2C8. Please note that this appendix is not an exhaustive list of all medications that may be substrates of CYP3A or CYP2C8. The sponsor should be contacted with any questions regarding concomitant use of medications that are CYP3A or CYP2C8 substrates.

6.5.2 Medications that are Substrates or Inhibitors of P-glycoprotein (for Patients Receiving Duvelisib Only)

In vitro data indicate that duvelisib is a substrate for P-glycoprotein (P-gp) and may have the potential to inhibit the activity of P-gp. P-gp substrates or inhibitors may be used as medically indicated but with caution.

[Appendix 5](#) provides a list of medications that are substrates or inhibitors of P-gp. Please note that this appendix is not an exhaustive list of all medications that may be substrates of P-gp or may modulate P-gp activity. The sponsor should be contacted with any questions regarding concomitant use of medications that are thought to modulate P-gp activity.

6.5.3 Medications that are Inhibitors of CYP1A2 (for Patients Receiving Bendamustine Only)

Bendamustine metabolism involves CYP1A2 isoenzyme. Therefore, the potential for interaction with CYP1A2 inhibitors such as fluvoxamine, ciprofloxacin, acyclovir, and cimetidine exists. Please refer to the bendamustine SmPC for further details.

6.6 Other Concomitant Therapies

Any other medication that is considered necessary for the patient's welfare, and that is not expected to interfere with the evaluation of duvelisib, may be given at the discretion of the investigator.

6.7 Photo Safety

Duvelisib was determined to have the potential for phototoxicity in an in vitro phototoxicity assay and in a follow-up in vivo phototoxicity study. Patients should be advised to use appropriate protective measures to minimise exposure to direct sunlight or ultraviolet light sources during the treatment period and for at least 30 days after the last dose of duvelisib.

Similar advice may be given to patients in the comparator arm according to institutional standards.

6.8 Blinding and Unblinding

Treatment assignment is not blinded.

6.9 Treatment Compliance

The investigator will dispense the study drug only for use by patients randomised in the study as described in this protocol. The study drug is not to be used for reasons other than those described in this protocol.

For patients receiving duvelisib, the investigator or other study staff will supervise study drug given in the clinic and instruct the patient on duvelisib self-administration. Patients will be asked to bring their duvelisib drug diary and blister cards with them at each visit. Compliance with protocol-defined duvelisib intake will be checked by drug diary and pill count. If the drug diary and blister card are not returned, the patient should be questioned regarding drug compliance during the study visit.

For patients receiving gemcitabine or bendamustine, suitable qualified and delegated site staff will administer the study drug in the clinic. Drug administration will be documented for accountability purposes.

7 DISCONTINUATION CRITERIA

7.1 Discontinuation/Withdrawal of Individual Patients from Study Drug

The study drug will be discontinued for any of the following reasons:

- A patient may withdraw from the study drug at any time at the patient's own request for any reason (or without providing any reason).
- Disease progression
- Clinical deterioration due to PD that does not meet Lugano 2014 criteria
- For patients receiving duvelisib: Study drug interruption for > 21 days due to duvelisib-related toxicity (unless approved by study medical monitor).
- AE, such as unacceptable toxicity
- Initiation of haematopoietic stem cell transplant
- Initiation of any other anti-cancer therapy
- Pregnancy
- Death
- Protocol Deviation / non-compliance
- Other reasons

The investigator must determine the primary reason for a duvelisib withdrawal and record this information on the eCRF. The patient will be observed after discontinuation according to the Schedule of Activities ([Table 1](#)).

7.2 Discontinuation/Withdrawal of Individual Patients from the Study

A patient may be discontinued from the study at any time for any of the following reasons:

- A patient may withdraw from the study at any time at the patient's own request for any reason (or without providing any reason).
- Termination of the study by the sponsor
- Death
- Lost to follow-up
- Other reasons, including major protocol violation and noncompliance with study procedures, as specified by the investigator

The investigator must determine the primary reason for a patient's withdrawal from the study and record this information on the eCRF.

7.3 Early Discontinuation of the Study

The sponsor may terminate this study at any time in accordance with the provisions outlined in Section 10.11.

7.4 Lost to Follow-up

A patient will be considered lost to follow-up if the patient repeatedly fails to return for scheduled visits and is unable to be contacted by the study site.

The following actions must be taken if a patient fails to return to the clinic for a required study visit:

- The site must attempt to contact the patient and reschedule the missed visit as soon as possible, counsel the patient on the importance of maintaining the assigned visit schedule, and ascertain whether the patient wishes to and/or should continue in the study.
- Before a patient is deemed lost to follow-up, the investigator or designee must make every effort to regain contact with the patient (where possible, 3 telephone calls, and if necessary, a certified letter to the patient's last known mailing address or local equivalent methods). These contact attempts should be documented in the patient's medical record.
- Should the patient continue to be unreachable, the patient will be considered lost to follow-up.

8 STUDY ASSESSMENTS

This study will be conducted in compliance with the protocol, with EU Regulation No 536/2014, with the principles of good clinical practice (GCP), other applicable regulatory requirements, and International Council for Harmonisation (ICH) guidelines.

8.1 Study Procedures

Patients will be evaluated at scheduled visits over several study periods: Screening phase, treatment phase, completion of treatment, safety follow-up phase, durability of response phase, and survival follow-up phase. Tests and procedures should be performed in accordance with the Schedule of Activities provided in [Table 1](#). Additional details are provided in the sections that follow.

8.1.1 Informed Consent

Each patient must sign a written informed consent form (ICF) prior to any study-specific procedure. Standard of care assessments that were performed prior to patient signing the ICF may be used at screening, provided they are performed within the screening window period.

8.1.2 Inclusion and Exclusion Criteria

All patients must meet the inclusion and exclusion criteria prior to being randomised in the study. Eligibility criteria will be reviewed during screening and again prior to dosing on C1D1.

8.1.3 Medical and Medication History

During screening, a complete medical history will be obtained from each patient including relevant medical history, current medications, current primary cancer diagnosis, prior cancer treatments (chemo- and immunotherapies, radiation therapy, surgeries) including duration of prior response, and disease-specific characteristics such as tumour stage and histology. Current medications and all medications used within 30 days prior to screening will be documented.

8.1.4 Disease Confirmation by Central Pathology Review

As part of the patient's medical history, pathology reports from the time of diagnosis or later that are relevant to the diagnosis are to be collected. In addition, stained formalin-fixed, paraffin-embedded (FFPE) slides used for nodal T cell lymphoma with TFH phenotype diagnosis should be collected for all screened patients and submitted for central pathology review. Refer to the laboratory manual for more details.

8.1.5 Demography

At screening, patient demographic data will be collected, including date of birth, gender, race, and ethnicity, as allowable according to local law.

8.2 Safety Assessments

8.2.1 Physical Examination

A complete physical examination per standard of care will be conducted at the screening. Targeted physical examinations should be performed at time points specified in the Schedule of Activities ([Table 1](#)).

Any physical examination abnormality deemed clinically significant by the investigator during screening will be reported as medical history. Any physical examination abnormality that emerges or has worsened after signing of the ICF and that is assessed as clinically significant by the investigator will be reported as an AE.

8.2.2 Electrocardiogram

A twelve-lead electrocardiogram (ECG) will be conducted during screening and assessed by the investigator to confirm eligibility. A single ECG tracing is sufficient unless there is an abnormality, such as prolonged QTcF ≥ 480 msec, new arrhythmia, or other clinically significant finding. If an abnormality is observed, the ECG is to be repeated at the investigator's discretion. Additional on-treatment or follow-up ECGs should be performed at the investigator's discretion as clinically indicated.

8.2.3 Vital Signs, Height, and Weight

Vital signs, including temperature ($^{\circ}\text{C}$), blood pressure, pulse, and respiration rate, as well as weight (kg) will be measured at the time points specified in the Schedule of Activities ([Table 1](#)). Height (cm) will be measured during screening only.

Systolic and diastolic blood pressure and pulse will be measured with the patient in a supine or sitting position; the same position will be used for a patient throughout the study. Blood pressure should be assessed on the same arm throughout the study.

8.2.4 ECOG Performance Status

Performance status will be assessed using the ECOG Scale (see [Appendix 6](#)) at the time points specified in the Schedule of Activities ([Table 1](#)).

8.3 Clinical Laboratory Tests

8.3.1 Screening Serology

Laboratory studies to assess the status of prior or current CMV infection, including CMV serology, CMV immunoglobulin (Ig) G and IgM, are to be conducted during screening. CMV PCR is to be performed as clinically indicated per investigator discretion. Patients with a detectable viral load will not be eligible for randomisation. During duvelisib treatment, periodic CMV viral load

monitoring is recommended in patients with a history of CMV reactivation or other opportunistic infections.

EBV serology and PCR can be performed at the investigator's discretion on an individual patient basis if considered medically relevant. There are no protocol-stipulated consequences for positive test results.

Furthermore, a liver serology panel, including HBsAg, HBcAb, and HCVAb, is to be performed during screening. Patients with a positive HBsAg or HCVAb in the absence of documented cure will be excluded. Patients with a positive HBcAb must have negative HBV DNA to be eligible, must receive prophylaxis with entecavir (or equivalent) concomitant with duvelisib treatment, and must be periodically monitored for HBV reactivation by institutional guidelines. Investigators who strongly believe that a positive HBcAb is false due to passive immunisation from previous immunoglobulin infusion therapy should discuss the potential to defer HBV prophylaxis with the medical monitor.

HIV are to be performed during screening for all patients without documentation of prior negative results within 2 years. Human T-lymphotropic virus type 1 (HTLV-1) only requires retesting if there is no documented negative test in the past.

8.3.2 Pregnancy Test

A serum pregnancy test will be done during screening and repeated pre-dose on C1D1 for WOCBP; results must be available and confirmed to be negative before administration of the first study drug dose. If the screening pregnancy test is performed within 3 days before C1D1, it needs not be repeated on C1D1. The pregnancy test may be repeated as deemed necessary by the investigator.

8.3.3 Clinical Laboratory Tests

Clinical laboratory tests will be performed by the investigator's site locally. Samples will be collected from patients at scheduled study visits before the administration of study drug unless otherwise noted.

Blood samples for evaluation of the following clinical laboratory tests will be collected at the time points specified in the Schedule of Activities ([Table 1](#)). Samples may be collected up to 3 days prior to the visit. C1D1 laboratory tests do not need to be repeated if performed within 3 days prior to C1D1. All safety screening/C1D1 laboratory results will be reviewed prior to dosing to ensure patient is eligible.

- Haematology: Haemoglobin, haematocrit, platelets, white blood cell count (WBC) with 5-part differential (neutrophils, lymphocytes, monocytes, eosinophils, basophils), such that an ANC and an absolute lymphocyte count (ALC) can be derived, if it is not already provided as part of the laboratory analysis.

- Clinical chemistry: Sodium, potassium, chloride, bicarbonate, albumin, total protein, creatinine, blood urea nitrogen or urea, lipase, amylase, uric acid, calcium, phosphorus, magnesium, glucose, lactic dehydrogenase, serum ALT, serum AST, bilirubin (total bilirubin may be reported; if total bilirubin is higher than normal range, both direct and indirect bilirubin should be reported), and alkaline phosphatase.
- CD4 lymphocyte count (at screening).

8.4 Disease and Response Assessment

At screening, disease will be measured using PET-CT (CT images with PET should be diagnostic with contrast wherever possible). Where PET-CT is not feasible, CT with contrast or MRI may be used. Subsequently, for FDG-avid PTCL, PET-CT will be used to assess disease status and for FDG-non-avid PTCL, CT with contrast or MRI will be appropriate substitutes. For each patient, only one of these modalities should be used to evaluate response, and the modality chosen for each individual patient should be the same throughout the duration of the study. In all cases, disease status will be assessed using the Lugano 2014 criteria ([Cheson 2014](#)).

PET images are to be assessed using the 5-point scale (5PS), a semi-quantitative analysis that is a pragmatic yet robust predictor of patient outcome ([Table 10](#)).

Table 10: 5-Point Scale (Lugano Classification)

Score	Criterion
1	No uptake above background
2	Uptake $\leq$ mediastinum
3	Uptake $>$ mediastinum, but $\leq$ liver
4	Uptake moderately $>$ liver*
5	Uptake markedly higher than liver and/or new lesions*
X	(New) areas of uptake unlikely to be related to lymphoma

*It is suggested according to published data that Score 4 be applied to uptake greater than the maximum standardised uptake value (SUV) in a large region of normal liver and Score 5 to uptake $2\times$ to $3\times$ greater than the maximum SUV in the liver.

Source: [Barrington 2014](#)

At each scheduled disease response assessment ([Table 1](#)), a PET-CT- or CT-based response will be determined, based on the Lugano 2014 response criteria for patients with malignant lymphoma ([Table 11](#)).

Table 11: Lugano 2014 Response Criteria for Malignant Lymphoma

Response and Site	PET-CT-based Response	CT-based Response
<i>Complete Response</i>	<i>Complete metabolic response</i>	<i>Complete radiologic response (requires all of the following)</i>
Lymph nodes and extralymphatic sites	Score of 1, 2, or 3* with or without a residual mass on 5PS It is recognised that in Waldeyer's ring or extranodal sites with high physiologic uptake or with activation within spleen or marrow (e.g., with chemotherapy or myeloid colony stimulating factors), uptake may be greater than normal mediastinum and/or liver. In this circumstance, complete metabolic response may be inferred if uptake at sites of initial involvement is no greater than surrounding normal tissue even if the tissue has high physiologic uptake	Target nodes/nodal masses must regress to ≤ 1.5 cm in LDi No extralymphatic sites of disease
Non-measured lesions	Not applicable	Absent
Organ enlargement	Not applicable	Regress to normal
New lesions	None	None
Bone marrow	No evidence of FDG-avid disease in bone marrow	Normal by morphology; if indeterminate, IHC negative
<i>Partial Response</i>	<i>Partial metabolic response</i>	<i>Partial remission (all of the following)</i>
Lymph nodes and extralymphatic sites	5PS score 4 or 5 with reduced uptake compared with baseline and residual mass(es) of any size During treatment, these findings suggest responding disease At end of treatment, these findings indicate residual disease	$\geq 50\%$ decrease in SPD of up to 6 target measurable nodes and extranodal sites When a lesion is too small to measure on CT, assign 5 mm $\times$ 5 mm as the default value When no longer visible, 0 $\times$ 0 mm For a node > 5 mm $\times$ 5 mm, but smaller than normal, use actual measurement for calculation
Non-measured lesions	Not applicable	Absent/normal, regressed, but no increase
Organ enlargement	Not applicable	Spleen must have regressed by $\geq 50\%$ in length beyond normal
New lesions	None	None

Response and Site	PET-CT-based Response	CT-based Response
Bone marrow	Residual uptake higher than uptake in normal marrow but reduced compared with baseline (diffuse uptake compatible with reactive changes from chemotherapy allowed). If there are persistent focal changes in the marrow in the context of a nodal response, consideration should be given to further evaluation with MRI or biopsy or an interval scan	Not applicable
Stable disease (no response)	No metabolic response	Stable disease
Target nodes/nodal masses, extranodal lesions	5PS score 4 or 5 with no significant change in FDG uptake from baseline at interim or end of treatment	< 50% decrease from baseline in SPD of up to 6 dominant, measurable nodes and extranodal sites; no criteria for progressive disease are met
Non-measured lesions	Not applicable	No increase consistent with progression
Organ enlargement	Not applicable	No increase consistent with progression
New lesions	None	None
Bone marrow	No change from baseline	Not applicable
Progressive Disease	Progressive metabolic disease	Progressive disease requires at least 1 of the following:
Individual target nodes/nodal masses	5PS score 4 or 5 with an increase in intensity of uptake from baseline and/or	PPD progression
Extranodal lesions	New FDG-avid foci consistent with lymphoma at interim or end of treatment assessment	An individual node/lesion must be abnormal with: LDi > 1.5 cm and Increase by $\geq 50\%$ from PPD nadir and An increase in LDi or SDi from nadir 0.5 cm for lesions ≤ 2 cm 1.0 cm for lesions > 2 cm In the setting of splenomegaly, the splenic length must increase by > 50% of the extent of its prior increase beyond baseline (e.g., a 15-cm spleen must increase to > 16 cm). If no prior splenomegaly, must increase by at least 2 cm from baseline New or recurrent splenomegaly
Non-measured lesions	None	New or clear progression of pre-existing non-measured lesions

Response and Site	PET-CT-based Response	CT-based Response
New lesions	New FDG-avid foci consistent with lymphoma rather than another aetiology (e.g., infection, inflammation). If uncertain regarding aetiology of new lesions, biopsy or interval scan may be considered	Regrowth of previously resolved lesions A new node > 1.5 cm in any axis A new extranodal site > 1.0 cm in any axis; if < 1.0 cm in any axis, its presence must be unequivocal and must be attributable to lymphoma Assessable disease of any size unequivocally attributable to lymphoma
Bone marrow	New or recurrent FDG-avid foci	New or recurrent involvement

5PS = 5-point scale; CT = computed tomography; FDG = fluorodeoxyglucose; IHC = immunohistochemistry; LD_i = longest transverse diameter of a lesion; MRI = magnetic resonance imaging; PET = positron emission tomography; PPD = cross product of the LD_i and perpendicular diameter; SD_i = shortest axis perpendicular to the LD_i; SPD = sum of the product of the perpendicular diameters for multiple lesions

*A score of 3 in many patients indicates a good prognosis with standard treatment, especially if at the time of an interim scan. However, in studies involving PET where de-escalation is investigated, it may be preferable to consider a score of 3 as inadequate response (to avoid undertreatment). Measured dominant lesions: Up to 6 of the largest dominant nodes, nodal masses, and extranodal lesions selected to be clearly measurable in 2 diameters. Nodes should preferably be from disparate regions of the body and should include, where applicable, mediastinal and retroperitoneal areas. Non-nodal lesions include those in solid organs (e.g., liver, spleen, kidneys, lungs), gastrointestinal (GI) involvement, cutaneous lesions, or those noted on palpation. Non-measured lesions: Any disease not selected as measured, dominant disease and truly assessable disease should be considered not measured. These sites include any nodes, nodal masses, and extranodal sites not selected as dominant or measurable or that do not meet the requirements for measurability but are still considered abnormal, as well as truly assessable disease, which is any site of suspected disease that would be difficult to follow quantitatively with measurement, including pleural effusions, ascites, bone lesions, leptomeningeal disease, abdominal masses, and other lesions that cannot be confirmed and followed by imaging. In Waldeyer's ring or in extranodal sites (e.g., GI tract, liver, bone marrow), FDG uptake may be greater than in the mediastinum with complete metabolic response but should be no higher than surrounding normal physiologic uptake (e.g., with marrow activation as a result of chemotherapy or myeloid growth factors).

Source: [Cheson 2014](#)

8.5 Exploratory Biomarkers

Plasma is to be collected at the times designated in the Schedule of Activities ([Table 1](#)) for exploratory biomarker analysis. In addition, whenever a bone marrow specimen is collected during screening, an aliquot is to be reserved for exploratory biomarker assessments. Each of the specimens listed may be analysed at a central laboratory or at a specialised laboratory vendor as determined by the sponsor, for the following biomarkers:

- AITL vs non-AITL
- Whole exome sequencing
- Circulating DNA
- AITL score, including B2M, CRP, and other parameters

8.6 Quality of Life Assessments

QoL will be obtained according to the Schedule of Activities ([Table 1](#)).

EORTC QLQ-C30

The European Organisation for Research and Treatment of Cancer (EORTC) QLQ-C30 is a validated, reliable self-report measure ([Aaronson 1993](#); [Fitzsimmons 1999](#)). It consists of 30 questions that assess 5 aspects of patient functioning (physical, emotional, role, cognitive, and social), 3 symptom scales (fatigue, nausea and vomiting, and pain), global health/QoL, and 6 single items (dyspnoea, insomnia, appetite loss, constipation, diarrhoea, and financial difficulties) with a recall period of the previous week (see [Appendix 7](#)). Scale scores can be obtained for the multi-item scales. The first 28 items are scored on a 4-point scale that ranges from “not at all” to “very much,” and the last 2 items are scores on a 7-point scale that ranges from “very poor” to “excellent.” Higher scores indicate higher response levels (i.e., higher health-related quality of life [HRQoL], higher symptom severity). The EORTC QLQ-C30 takes approximately 10 minutes to complete. It will be utilised in this study for informing overall HRQoL evaluation.

EQ-5D-5L

The EQ-5D-5L is a validated self-report health status questionnaire that is used to calculate a health status utility score for use in health economic analyses ([EuroQol Group 1990](#); [EuroQol 2018](#); [Brooks 2020](#); [Herdman 2011](#); [Janssen 2013](#)). There are 2 components to the EQ-5D-5L: a five-item health state profile that assesses mobility, self-care, usual activities, pain or discomfort, and anxiety or depression, and a visual analogue scale that measures overall health state (see [Appendix 8](#)). Published weighting systems allow for creation of a single composite score of the patient’s health status. The EQ-5D-5L takes approximately 3 minutes to complete. It will be utilised in this study for informing pharmacoeconomic evaluations.

EORTC QLQ-NHL-HG29

The EORTC QLQ-NHL HG 29 is a validated disease-specific QoL questionnaire to supplement the EORTC QLQ-C30 and was developed for patients with high-grade NHL ([van de Poll-Franse 2018](#); [Oerlemans 2023](#)). It consists of 29 questions contributing to 5 multi-item subscales and 3 conditional items (see [Appendix 9](#)). The subscales include symptom burden due to disease and/or treatment (7 items), neuropathy (2 items), physical condition/fatigue (5 items), emotional impacts (4 items), and worries about health and functioning (8 items). The 3 conditional items, which patients complete only if relevant to them, are about having problems at work/education, worries about work/education, and concerns about the ability to have children. Items are rated using a 4-point response scale (“not at all,” “a little,” “quite a bit,” and “very much”) and the reference time frame for all items is the past week. The EORTC QLQ-NHL-HG29 takes approximately 10

minutes to complete. It will be utilised in this study for informing disease-specific HRQoL evaluation.

8.7 Unscheduled Visits

Unscheduled visits and assessments may occur at the investigator's discretion, as clinically warranted.

8.8 Concomitant Medications

Medications received on or after the start of duvelisib therapy through the Safety Follow-up visit will be documented.

8.9 Adverse Event Management

8.9.1 Definition of an Adverse Event

An AE is defined as any untoward medical occurrence in a patient administered a medicinal product that does not necessarily have a causal relationship with this treatment. An AE can, therefore, be any unfavourable and unintended sign (including an abnormal laboratory finding), symptom, or disease temporally associated with the use of a medicinal (investigational) product, whether or not related to the study (investigational) product. This includes an exacerbation of pre-existing conditions or events, concurrent illnesses, drug interaction, or the significant worsening of the indication under investigation. Anticipated fluctuations of pre-existing conditions, including the disease under study that does not represent a clinically significant exacerbation or worsening, need not be considered AEs.

Symptoms of the disease under study, lack of efficacy, or disease progression should not be classified as an AE as long as they are within the normal day-to-day fluctuation or expected progression of the disease.

It is the responsibility of the investigator to document all AEs that occur during the study. AEs should be reported on the appropriate page of the eCRF.

8.9.2 Definition of a Serious Adverse Event

An SAE is any untoward medical occurrence at any dose (including after the ICF is signed and prior to dosing) that:

- Results in death
- Is life-threatening (patient is at immediate risk of death from the event as it occurred)
- Requires inpatient hospitalisation (formal admission to a hospital for medical reasons) or prolongation of existing hospitalisation
- Results in persistent or significant disability/incapacity

- Results in a congenital anomaly/birth defect

Important medical events that may not result in death, are not immediately life-threatening, or do not require hospitalisation may be considered SAEs when, based on appropriate medical judgement, they may jeopardise the patient and may require medical or surgical intervention to prevent one of the outcomes listed in this definition. Examples of such events include allergic bronchospasm requiring intensive treatment in an emergency room or at home, blood dyscrasias or convulsions that do not result in inpatient hospitalisation, or the development of drug dependency or drug abuse.

For SAE reporting purposes, hospitalisation is defined as an inpatient hospital stay ≥ 24 hours. Hospitalisations for elective surgery or other medical procedures that are not related to a TEAE are not considered SAEs. Hospitalisation, which in the opinion of the investigator, is unrelated to the study drug, and due to purely non-medical circumstances (e.g., respite care, lack of a caretaker at home, lack of transportation home) are also not considered to be SAEs. PD under study (including signs and symptoms of progression) if documented by use of appropriate methods, should not be reported as an SAE unless the outcome is fatal during the study or within the safety reporting period (see Section 8.9.3). If the malignancy has a fatal outcome during the study or within the safety reporting period, then the event should be reported using the term “disease progression” with a CTCAE severity of Grade 5.

Death is an outcome. The primary reason for patients’ death should be reported as an SAE.

8.9.3 Reporting of Adverse Events and Serious Adverse Events

The AE reporting period begins from the time that the patient signs the ICF through and including 30 calendar days after the last study drug dose. All treatment-related AEs/SAEs should be followed until resolution or stabilisation.

Investigators are not obligated to actively seek information on AEs or SAEs after the Safety Follow-up visit (i.e. during the period of Durability of Response and Survival Follow-up). However, if the investigator learns of any SAE, including a death, at any time later than 30 (+7) days after last dose of study drug, and the investigator considers the event to be reasonably related to the study drug, the investigator must promptly notify the sponsor.

Elective or previously scheduled hospitalisations for pre-existing conditions that have not worsened after initiation of treatment should not be classified as SAEs. For example, an admission for a previously scheduled ventral hernia repair would not be classified as an SAE.

If medically plausible, the condition, disease, or syndrome should be reported rather than each individual sign or symptom. If a constellation of signs and/or symptoms cannot be medically characterised as a single diagnosis or syndrome at the time of reporting, each individual event should be recorded as an AE or SAE as appropriate on the relevant form(s) (SAE Report Form and/or AE eCRF). If a diagnosis is subsequently established, it should be reported as follow-up

information is available. If a diagnosis is determined subsequent to the reporting of the constellation of symptoms, the signs/symptoms should be updated to reflect the diagnosis.

Each AE is to be evaluated for duration, severity, seriousness, and causal relationship to the investigational drug. The action taken with study drug and the outcome must also be recorded.

All SAEs, regardless of relationship to the study treatment, must be reported to the sponsor and Contract Research Organization (CRO) pharmacovigilance group, without undue delay but not later than within 24 hours of obtaining knowledge of the events. Initial SAE notification should be made by e-mailing the SAE report form to the e-mail provided on the SAE report form.

An initial SAE report may be sent without the investigator's signature but must be followed by a report signed by the investigator within 48 hours of becoming aware of the event. Follow-up SAE reports must be submitted by the investigator as new information becomes available.

The medical monitors for this study may be contacted in case of need for advice or assistance. Refer to Investigator Site File for Contact details.

8.9.4 Severity of Adverse Events

The severity of the AE will be graded according to the NCI-CTCAE version 5.0 (see web page at <http://ctep.cancer.gov> for details). AEs not listed in the CTCAE should be graded as summarised in [Table 12](#).

Table 12: CTCAE Grading

CTCAE Grade	Equivalent To	Definition
Grade 1	Mild	Discomfort noticed but no disruption of normal daily activity
Grade 2	Moderate	Discomfort sufficient to reduce or affect daily activity; no treatment or medical intervention is indicated although this could improve the overall well-being or symptoms of the patient
Grade 3	Severe	Inability to work or perform normal daily activity; treatment or medical intervention is indicated in order to improve the overall well-being or symptoms; delaying the onset of treatment is not putting the survival of the patient at direct risk
Grade 4	Life-threatening/ disabling	An immediate threat to life or leading to a permanent mental or physical conditions that prevents work or performing normal daily activities; treatment or medical intervention is required in order to maintain survival
Grade 5	Death	AE resulting in death

8.9.5 Relationship of Adverse Events to Study Drug

The investigator is obligated to assess the relationship to study drug for each occurrence of each AE/SAE. The categories for classifying the investigator's opinion of the relationship are as follows:

- Related: A reasonable possibility exists of a relationship between the AE and the IP.
- Not related: No reasonable possibility exists of a relationship between the AE and the IP.

“Reasonable possibility” means there is evidence to suggest a causal relationship between the drug and the event. Factors that should be considered when making a causality assessment include, but are not limited to, temporal relationship of the event to drug administration; biologic plausibility, based on the mechanism of action of the drug or similar drugs in the same class; nonclinical evidence; dechallenge/rechallenge information. The anticipated occurrence of the event in a patient population unexposed to the study drug is also a factor to be considered.

8.9.6 Clinical Laboratory Adverse Events

A clinical laboratory AE is any laboratory value that is deemed clinically significant by the investigator, is greater than Grade 2 in severity, and is accompanied by one of the following:

- requires a medical intervention
- requires a change or suspension of study drug
- is accompanied by clinical symptoms

Laboratory abnormalities that have not required medical intervention should not be recorded as AEs and will be captured and reported in the laboratory section of the clinical study report. If a medical intervention occurs, it should be recorded as a treatment with the abnormal laboratory finding as the AE (e.g., anaemia with treatment required and blood transfusion recorded as a procedure, hyperglycaemia with treatment required and change in insulin dose recorded on concomitant medications).

The investigator should decide, based upon the AE criteria and the clinical condition of the patient, whether a change in a laboratory parameter is clinically significant and therefore represents an AE.

If, at the end of the treatment phase with the study drug, there are pathological laboratory values which were not present at baseline, further clinical or laboratory investigations should be performed until the values return to within reference range or until a plausible explanation (i.e., concomitant disease) is found for the pathological laboratory values.

8.9.7 Regulatory Aspects of Adverse Event Reporting

The sponsor is responsible for submitting reports of suspected unexpected serious adverse reactions associated with the use of the study drug to the appropriate regulatory authority, investigators, and IEC/CEC in accordance with all applicable regulations and guidelines.

The investigator is obligated to learn about the safety reporting requirements for investigators in their country. Each investigator is required to comply with applicable regulations (including local laws and guidance) regarding the notification of safety reports to their health authorities, IRB/IEC, coordinating investigators, study investigators, and institutions.

8.9.8 Overdose

For this study, duvelisib overdose is defined as a daily dose of duvelisib higher than the prescribed daily dose. Overdoses will not be considered SAEs unless the outcome of the overdose meets seriousness criteria as defined in Section 8.9.2. In the event of an overdose, the sponsor should be immediately notified. The patient should be carefully monitored for potential adverse reactions and symptomatic treatment instituted as per institutional standards of care.

In the case of duvelisib overdose, clinic staff should be notified immediately, and supportive care is to be given as indicated. Patients should be informed to contact their doctor immediately if they have taken an overdose and should stop taking duvelisib.

In case of comparator overdose, please refer to the respective SmPC.

8.9.9 Pregnancy

Pregnancies occurring in patients or partners of male patients after the study participant's ICF is signed up until 30 days after the patient's last dose of study treatment must be reported to the sponsor or designee within 24 hours of the investigator's knowledge of the pregnancy using a Pregnancy Report Form by e-mailing the form to the e-mail address provided on that form. If a pregnancy occurs in a patient, study treatment must be discontinued immediately. The pregnant woman should be referred to an obstetrician/gynaecologist experienced in reproductive toxicity for further evaluation and counselling. The pregnant partner of the participant will be asked to provide consent to share their health information with the investigator.

Pregnancy per se is not considered an AE unless there is cause to believe that the study drug may have interfered with the effectiveness of a contraceptive medication or if the outcome of the pregnancy meets SAE criteria (miscarriage or congenital anomaly/birth defect, etc.), in which case it should be reported in the same manner and timelines as an SAE. In addition, any infant death or congenital anomaly occurring after 30 days that the investigator suspects is related to the in-utero exposure to the study drug should also be reported as an SAE. Hospitalisation for normal delivery of a healthy newborn is not an SAE.

Pregnant women and WOCBP who are not using effective contraception are excluded from the study (see Section 8.3.2 and Section 6.3.3 for instructions on pregnancy testing and birth control).

The investigator will follow the pregnant woman until completion of the pregnancy, provided that the pregnant woman consents to pregnancy follow-up, and must notify the sponsor of the outcome within 24 hours of the investigator's knowledge of the pregnancy outcome using a Pregnancy Outcome Form. This notification includes pregnancies resulting in live, "normal" births.

9 STATISTICAL CONSIDERATIONS

9.1 Statistical Analysis Plan

The detailed methodology for summary and statistical analyses of the data collected in this study will be documented in a Statistical Analysis Plan. The Statistical Analysis Plan will provide all data handling rules, including the management of missing values and the handling of data for withdrawn patients. The Statistical Analysis Plan will also outline protocol deviation criteria. Any deviations from the planned analyses specified or populations defined within the Statistical Analysis Plan will be justified in writing and presented within the final clinical study report.

The clinical database lock will occur after all study data are reconciled (i.e., “cleaned”) for all patients who have participated in the study. At the time of the interim analysis, the data will be cleaned for all relevant variables for the time-to-event endpoint of PFS and then the data base will be frozen. The Statistical Analysis Plan for the study will be finalised and signed before the database lock.

9.2 Determination of Sample Size and Power Considerations

9.2.1 Primary Analysis

This study employs an event-driven, group sequential design. The planned number of PFS events at 30 months after enrolment of first participant is estimated to be 108. This will provide approximately 90% power to detect a difference in PFS if the true HR of Arm 1 and Arm 2 is 0.53 (5.3 months vs 2.8 months in median PFS, respectively), based on a 1-sided stratified log-rank test controlling the type 1 error rate at 0.025.

One interim analysis for efficacy and futility is planned, which will be conducted at approximately 50% of the planned number of PFS events. The boundaries for efficacy and futility at the interim analysis, will be the Hwang-Shih-DeCani alpha and beta spending functions with spending parameters $\gamma = -4$ (approximate O’Brien-Fleming design) and $\gamma = -2$, respectively. Assuming a 10% annual drop-out rate with drop-out time exponentially distributed, the required sample size is approximately 124 patients. With patient accrual of 0.09 patients per site per month, accrual of patients is projected to last approximately 30 months, and the study duration is expected to be approximately 48 months.

9.2.2 Power Consideration Key Secondary Analysis

The estimated number of OS events at 48 months after enrolment of the first patient is approximately 92. This will provide 80% power to detect a difference in OS if the true HR for Arm 1 and Arm 2 is 0.55 (18 months vs 10 months in median OS, respectively), based on a 1-sided stratified log-rank test controlling the type 1 error rate at 0.025.

Sample size calculation have been done in R using package rpact version 3.5.0.

9.3 Randomisation and Stratification

Patients will be randomly assigned on C1D1 to either Arm 1 or Arm 2 in a 1:1 ratio via a computer-generated randomisation scheme. Randomisation will be stratified based on AITL score (0-2 vs 3-4) and number of prior lines of therapy for T cell lymphoma (1 vs >1).

9.4 Populations for Analysis

Patients who were screened and signed the informed consent but were not randomised will be listed including reason for screen failure and any SAE that is related to study procedures. These patients will not be part of any summary table except for summarising disposition.

The following analysis sets will be defined:

- The **full analysis set** (FAS) consists of all randomised patients, which is the same as all patients to whom study treatment has been assigned. Following the intent-to-treat (ITT) principle, patients are analysed according to the treatment and stratum to which they were assigned at randomisation. The FAS will be the primary population for all efficacy-related endpoints.
- The **safety set** consists of all patients who received at least 1 dose of study medication. The safety set will be used for the safety analyses. Patients will be analysed as treated.

All patients randomised into the study will be included in data listings. Further details will be provided in the Statistical Analysis Plan.

All available efficacy and safety data will be included in data listings and tabulations. Data that are potentially spurious or erroneous will be examined according to standard data management operating procedures.

In general, missing data will be treated as missing and no data imputation will be applied, unless otherwise specified. Censoring rules are defined in Section [9.4.5](#).

9.4.1 Patient Disposition

The number of screened patients, screen failures, treated patients, patients who completed the study, and patients who discontinued the study will be summarised. Reasons for discontinuation will also be summarised.

9.4.2 Demographic and Baseline Characteristics

Demographic and baseline characteristics will be summarised in a descriptive fashion. Data to be tabulated will include gender, age, race, ethnicity, height, and weight, as well as disease-specific characteristics and other parameters as appropriate. Medical history, prior medication and treatments including prior anti-cancer therapies and surgeries data will be listed and summarised using descriptive statistics for numerical data and contingency tables for categorical data. All prior

therapies will be coded using World Health Organization Anatomical, Therapeutic and Chemical (WHO-ATC) terms.

9.4.3 Concomitant Medication and Therapy

Concomitant medication and significant non-drug therapies after the start of study treatment will be listed and summarised by WHO-ATC term in contingency tables.

9.4.4 Exposure

Treatment exposure will be summarised by cycle and overall.

9.4.5 Efficacy Analyses

Hypothesis for the Primary Objective:

Null Hypothesis: $H_0: ST_1(t) = ST_2(t)$

There is no difference in the survival functions of duvelisib vs. gemcitabine or bendamustine.

Vs.:

Alternative Hypothesis: $H_1: ST_1(t) \neq ST_2(t)$

There is a difference in the survival functions of duvelisib vs. gemcitabine or bendamustine.

Where:

- $ST_1(t)$ is the survival function for duvelisib at time t , representing the probability that a subject who received duvelisib survives beyond time t .
- $ST_2(t)$ is the survival function for gemcitabine or bendamustine at time t , representing the probability that a subject who received gemcitabine or bendamustine survives beyond time t .

Primary Endpoint Analyses of PFS

The primary endpoint is PFS, defined as the time from randomisation to documented PD, according to the Lugano 2014 criteria as assessed by the IRC, or death due to any cause. Patients without these events by the analysis data cut-off will be censored. The censoring rules for PFS are presented in [Table 13](#).

Table 13: Censoring Rules for PFS

Situation	Date of Event or Censoring	Event or Censoring
No adequate baseline assessment	Date of randomisation	Censored
No post-baseline assessment		
Progression	Date of progression	Event
Death	Date of death	Event

No progression (or death)	Date of last assessment	Censored
Progression or death after 2 or more missing assessments	Date of last assessment prior to missed assessments	Censored
Subsequent anti-cancer therapy administered	Date of last assessment prior to subsequent anti-cancer therapy	Censored
Stem cell transplantation (SCT)	Date of SCT	Censored

The primary PFS analysis of the study will apply a 1-sided stratified log-rank test for comparison of Arm 1 (duvelisib) and Arm 2 (gemcitabine or bendamustine) with Hwang-Shih-DeCani alpha spending functions with parameter $\gamma = -4$ and overall significance level 2.5%. The stratification factor will be the same as for randomisation (i.e. AITL score (0-2 vs 3-4) and number of prior lines of therapy for T cell lymphoma (1 vs >1)). The treatment effect will be quantified via a HR computed from a stratified Cox proportional-hazards regression, including a 2-sided 95% CI. Kaplan-Meier (KM) curves will be presented for each treatment arm. Median PFS and its 95% CI will be provided. In addition, PFS rates at month 2, 4, 6 and 8 and the associated 2-sided 95% CI will be calculated using the KM method.

The following sensitivity analyses of the primary endpoint are planned in case there is more than 10% of such events:

- Study drug discontinuation for Clinical deterioration due to PD that does not meet Lugano 2014 criteria: Considered as an event
- Study drug discontinuation for toxicity or other reason: Considered as event
- Start of a new anti-cancer treatment: Considered as event
- Missed 2 or more assessments (visits): Censoring at last available assessment before 2 or more missed assessments

Interim and Final PFS Analysis

Two PFS analyses are planned for this study. The first analysis is an interim analysis when approximately 50% of events have occurred to assess efficacy and futility (non-binding). The overall type 1 error rate will be controlled at a 1-sided 0.025 level through fixed alpha spending. Based on the expected information fraction available at this time, [Table 14](#) presents the expected stopping boundaries for the interim analyses and the final analysis for PFS. Exact efficacy and futility boundaries will be re-calculated based on the observed proportion at the time of this analysis. If 37 events (40% of the planned 92) for overall survival have been observed at the final analysis for PFS, a futility analysis for OS will be conducted. Otherwise, OS will be presented descriptively in case 36 or less sufficient events for overall survival have been collected.

Table 14: Expected Stopping Boundaries for Interim and Final PFS Analyses

Stage	Number of Events	HR at Boundary Efficacy / Futility	p (1-sided) Efficacy
Interim at 50% of events (expected) N=84	54	0.472 / 0.894	0.0032
Final (expected) N=124	108	0.682	0.0238

Subgroup Analyses

Exploratory subgroup analyses of the primary endpoint may be performed based on the stratification factors, treatment choice in the comparator arm (gemcitabine or bendamustine), age, and type of prior treatment regimen. Details on the subgroup analysis, including potential other factors, will be provided in the Statistical Analysis Plan.

In addition, outcomes associated with AITL vs non-AITL morphology are planned.

Key Secondary Endpoint Analyses

The OS analysis will be conducted once all patients have completed the follow-up period. OS is defined as the time from randomisation to the date of death due to any cause. If a death has not occurred by the analysis cut-off date, OS will be censored at the date of last contact.

For the comparison of duvelisib and gemcitabine or bendamustine, an OS analysis will apply a one-sided stratified log-rank test. The stratification factors will include AITL score (0-2 vs 3-4) and the number of prior lines of therapy for T cell lymphoma (1 vs >1). This test is designed to control the overall type 1 error rate at 0.025. However, OS will only be formally tested if the primary endpoint PFS has shown significance.

The treatment effect will be quantified using a HR calculated from a stratified Cox proportional-hazards regression, along with a 95% CI. To visually display the data, KM curves will be presented for each treatment arm.

Summary statistics for OS will be presented in tabular format for each dose cohort using KM methodology. This will include estimated median OS (in months) with 95% CI, 25th, and 75th percentiles. Additionally, KM estimated OS probabilities at timepoints of at least 2, 4, 6, 8, 10, 12, 18, 24, and 36 months will be calculated, along with corresponding 95% CIs. The tables will also include the number of patients at risk and the number of patients who experienced events at each timepoint.

Secondary Efficacy Endpoint Analyses

The secondary endpoints to be analysed are defined in Section 3. All secondary endpoint analyses will be performed in an exploratory fashion.

DOR

The DOR defined as time from first assessment of PR or CR to follow-on first assessment of PD will be summarised by descriptive statistics including median duration of response and respective 95% CIs. The same censoring rules as for the primary endpoint PFS will be used. DOR will also be listed.

ORR, CRR, and Proportion of Patients Who Proceed to SCT

ORR, CRR, and proportion of patients who proceed to SCT will be compared between the 2 treatment arms using a Cochran-Mantel-Haenszel test stratified by the randomisation stratification factor. In addition, proportions and 95% CI will be provided for each treatment arm.

PFS in Patients Who Proceed to SCT

PFS in patients who proceed to SCT will be analysed in the same way as described for the primary endpoint except for the censoring rule for SCT, which will not be applied.

QoL

QoL, including observed value and change from C1D1, will be tabulated by treatment group and time point in the study. Difference in change from C1D1 will be examined using an analysis of covariance model for repeated measures that will include the main effect of treatment and covariates and interaction of time and treatment. Differences and 95% CI in least square means will be reported for each time point.

9.4.6 Safety Analyses

Safety analyses will be performed on the Safety set, and safety endpoints will be tabulated and presented.

Adverse Events

AEs, study drug-related AEs, SAEs, study drug-related SAEs, AEs with NCI-CTCAE v5.0 $\geq$ Grades 3, AEs leading to premature discontinuation, interruptions, duration of interruptions, discontinuation of study treatment, and Standardised Medical Dictionary for Regulatory Activities (MedDRA) Queries (SMQs) will be analysed descriptively utilising corresponding MedDRA, system organ class, and preferred term. NCI-CTCAE v5.0 toxicity grades will be utilised for classifying severity.

Deaths within 30 days after the last dose and the corresponding reasons will be summarised. Also, all deaths overall will be summarised.

Clinical Laboratory Values

Safety laboratory results will be graded by NCI-CTCAE version 5 (or higher); if no grading exists, values will be classified into low/normal/high based on laboratory normal ranges. Each parameter

will be presented by descriptive statistics at each visit including change from baseline. Shift tables for NCI-CTCAE grades and normal ranges will be presented. All laboratory values will be listed. A separate listing for abnormal laboratory values (Grade 3 and higher, and low/high values) will be presented.

Vital Signs

Vital signs will be summarised by descriptive statistics at each visit including change from baseline. A listing will be provided.

9.4.7 Biomarker Data

Biomarkers will be evaluated with regard to baseline levels and change to baseline levels. Additional exploratory analyses identified by the sponsor also may be performed. Biomarker effects will be summarised using descriptive statistics and associations with clinical efficacy and/or safety outcomes may be explored.

10 ADMINISTRATIVE REQUIREMENTS

10.1 Good Clinical Practice

The study will be conducted in accordance with ICH GCP and the appropriate regulatory requirement(s). The investigator will be thoroughly familiar with the appropriate use of the study drug as described in the protocol and the Investigator's Brochure.

10.2 Ethical Considerations

Before implementing this study, the protocol, the proposed patient informed consent forms, and any other information provided to the patients, must be reviewed and approved by a properly constituted IRB/IEC/CEC.

Any substantive change or addition to the protocol can only be made in a written protocol amendment that must be approved by the sponsor, the IRB/IEC/CEC, and the regulatory authority.

The study will be conducted only at sites where IRB/IEC/CEC approval has been obtained. The protocol, Investigator's Brochure, ICF, advertisements (if applicable), written information given to the patients (including diary cards), safety updates, annual progress reports, and any revisions to these documents will be provided to the IRB/IEC by the investigator or the sponsor, as required by local regulations.

10.3 Informed Consent

After the study has been fully explained, written informed consent will be obtained from either the patient or his/her guardian or legal representative prior to study participation. The method of obtaining and documenting the informed consent and the contents of the consent must comply

with ICH GCP and all applicable regulatory requirements. Patients who are detained by administrative or judicial order will be excluded from participating in the study.

10.4 Patient Confidentiality and Disclosure

The investigator and investigator's staff shall maintain the confidentiality of all patient records. To maintain patient privacy, all eCRFs, study drug records, reports and communications will identify the patient by initials where permitted and/or the assigned patient number. The patient's confidentiality will be maintained and will not be publicly available to the extent permitted by the applicable laws and regulations.

10.5 Data Collection

The investigator is required to prepare and maintain adequate and accurate case histories designed to record all observations and other data pertinent to the study for each patient.

Secura Bio or designee will provide the study sites with secure access to and training on the electronic data capture system, sufficient to permit site personnel to enter or correct information in the eCRFs. eCRFs will be completed for each study patient. The investigator is responsible for ensuring the accuracy, completeness, and timeliness of the data entered in the eCRF. The audit trail will indicate the user's identification information and date and time of entry and/or change.

eCRFs should be completed in a timely manner, and every effort should be made to have forms completed and up to date.

At the conclusion of the study, Secura Bio or a designee will retain the eCRF data and corresponding audit trails. A copy of the final archival eCRF in the form of a CD or other electronic media will be placed in the investigator's study file.

10.6 Patient Data Protection

Patients will be assigned a unique identifier during screening. Any patient records or datasets that are transferred to the sponsor will contain the identifier only; patient names or any information that would make the patient identifiable will not be transferred.

The patient must be informed that his/her personal study-related data will be used by the sponsor in accordance with local data protection law. The level of disclosure must also be explained to the patient who will be required to give consent for their data to be used as described in the informed consent.

The patient must be informed that his/her medical records may be examined by Clinical Quality Assurance auditors or other authorised personnel appointed by the sponsor, by appropriate IRB/IEC/CEC members, and by inspectors from regulatory authorities.

The contract between sponsor and study sites specifies responsibilities of the parties related data protection, including handling of data security breaches and respective communication and cooperation of the parties. Information technology systems used to collect, process, and store study-related data are secured by technical and organisational security measures designed to protect such data against accidental or unlawful loss, alteration, or unauthorised disclosure or access.

10.7 Data Quality Assurance and Monitoring

The investigator must permit study-related monitoring, audits, IRB/IEC review, and regulatory agency inspections and provide direct access to source data documents.

Monitoring details describing strategy (e.g., risk-based initiatives in operations and quality such as Risk Management and Mitigation Strategies and Analytical Risk-Based Monitoring), methods, responsibilities, and requirements (including handling of noncompliance issues and monitoring techniques [central, remote, or on-site monitoring]) are provided in the Monitoring Plan.

The sponsor or designee is responsible for the data management of this study, including quality checking of the data.

The sponsor assumes accountability for actions delegated to other individuals (e.g., contract research organisations).

Study monitors will perform ongoing source data verification to confirm that data entered into the eCRF by authorised site personnel are accurate, complete, and verifiable from source documents; that the safety and rights of patients are being protected; and that the study is being conducted in accordance with the currently approved protocol and any other study agreements, ICH GCP, and all applicable regulatory requirements. Definition of what constitutes source data can be found in the Clinical Monitoring Plan.

Upon study completion, the monitor will arrange for a final review of the study files, after which the file should be secured by storage for the appropriate period as specified in Section [10.12](#).

10.8 Auditing Procedures

In addition to routine monitoring Secura Bio or designee, the IRB/IEC/CEC, or regulatory authority may inspect the study (during the study or after its completion) to evaluate compliance with the protocol and GCP.

The investigator agrees that representatives of the sponsor and regulatory authorities will have direct access, both during and after the course of this study, to audit and review all study-relevant records.

10.9 Investigator Compliance

The investigator will conduct the study in compliance with the protocol provided by Secura Bio and approved by the IRB/IEC/CEC and the appropriate regulatory authority. Modifications to the protocol will not be made without agreement of Secura Bio, the IRB/IEC/CEC and the regulatory authority, except where modification is needed to eliminate an immediate hazard to the patient. In this event, the investigator will contact Secura Bio or designee to discuss the planned course of action. Any deviation from the protocol must be documented.

10.10 Product Complaints

If a product complaint indicating dissatisfaction regarding duvelisib is received, the investigator should contact Secura Bio or designee. For product complaints regarding the comparators, the investigator should contact the respective manufacturer.

A product complaint in and of itself is not an AE. If a product complaint results in an SAE, an SAE form should be completed per the defined procedure.

10.11 Study Closure

Secura Bio or designee will notify the regulatory authorities and IRB/IEC/CECs that the study has ended.

Study participation by individual sites or the entire study may be prematurely terminated if in the opinion of the Investigator or Secura Bio there is sufficient reasonable cause. Written notification documenting the reason for study termination will be provided to the Investigator or Secura Bio by the terminating party.

Circumstances that may warrant termination include but are not limited to:

- Determination of unexpected, significant, or unacceptable risk to patients
- Failure to enter patients at an acceptable rate
- Insufficient adherence to protocol requirements
- Insufficient, incomplete, and/or unevaluable data
- Plans to modify, suspend, or discontinue the development of the study drug

10.12 Retention of Documents

Records and documents, including signed ICFs, pertaining to the conduct of this study must be retained by the investigator for 25 years after study completion unless local regulations or institutional policies require a longer retention period. No records may be destroyed during the retention period without the written approval of the sponsor. No records may be transferred to another location or party without written notification to the sponsor. If the investigator withdraws

from the responsibility of keeping the study records, custody must be transferred to a person willing to accept the responsibility and Secura Bio notified.

10.13 Disclosure of Information

All information provided to the investigator by Secura Bio, Inc. or its designee, will be kept strictly confidential. The investigator agrees to use this information to accomplish the study and will not use it for other purposes without consent from Secura Bio. No information about this study or its progress will be provided to anyone not involved in the study other than to Secura Bio or its designees, or in confidence to the IRB, or similar committee, except if required by law.

Upon completion of the clinical study and evaluation of results by Secura Bio, the hospital or institution, and/or investigator may publish or disclose the clinical study results pursuant to the terms contained in the applicable Clinical Study Agreement. Sponsor will comply with current regulatory requirements for disclosure and submission of study results.

The results of this study may be published or presented at scientific meetings with prior approval of the sponsor. The investigator agrees to submit all manuscripts or abstracts to the sponsor before submission. This allows the sponsor to protect proprietary information and to provide comments.

The sponsor will comply with the requirements for publication of study results. In accordance with standard editorial and ethical practice, the sponsor will generally support publication of multi-site studies only in their entirety and not as individual site data. In this case, a coordinating investigator will be designated by mutual agreement between the sponsor and the investigators.

Authorship will be determined by mutual agreement and in line with International Committee of Medical Journal Editors authorship requirements.

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12 APPENDICES

APPENDIX 1: ANGIOIMMUNOBLASTIC T CELL LYMPHOMA (AITL) SCORE

Description	Points
Age $\geq$ 60 years	1
ECOG performance status $>2^*$	1
$\beta 2$ microglobulin $>$ ULN	1
CRP $>$ ULN	1
* Maximum AITL score is 4, but in this study, there is an inclusion criterion of ECOG score ≤ 2 Source: Advani 2021	

Reference:

Advani RH, Skrypets T, Civaliero M, et al. Outcomes and prognostic factors in angioimmunoblastic T-cell lymphoma: final report from the international T-cell project. Blood. 2021;138(3):213-220.

APPENDIX 2: CONTRACEPTION GUIDANCE

The effects of duvelisib on conception, pregnancy, and lactation are unknown. Every woman of childbearing potential (WOCBP) and male patients with a partner who is a WOCBP must use contraception as described below. A woman is considered to be a WOCBP after menarche and until becoming postmenopausal (i.e., > 55 years and postmenopausal for at least 1 year), unless permanently sterile.

At screening, all male and female patients of reproductive potential (i.e., not surgically sterile, or female patients who are not postmenopausal) must agree to use a highly effective method of contraception (see below) for the duration of study treatment, and for at least 3 months after the last dose of duvelisib. Male patients must also refrain from donating sperm during their participation in the study and for at least 3 months after the last dose of duvelisib.

The use of contraceptive methods is not required if the male patient or the male partner (of the female patient) has a documented history of a vasectomy or if the female patient or the female partner (of the male patient) has a documented history of bilateral oophorectomy, hysterectomy, or tubal ligation, or if she is > 55 years of age and postmenopausal for at least 1 year.

See Section 8.9.9 for requirements for pregnancy reporting.

Highly Effective Forms of Contraception

Methods that can achieve a failure rate of less than 1% per year when used consistently and correctly are considered as highly effective birth control methods.

Such methods include:

- Combined (oestrogen and progestogen containing) hormonal contraception associated with inhibition of ovulation^a:
 - Oral
 - Intravaginal
 - Transdermal
- Progestogen-only hormonal contraception associated with inhibition of ovulation^a
 - Oral
 - Injectable
 - Implantable^b
 - Intrauterine device (IUD)^b
 - Intrauterine hormone-releasing system (IUS)^b
 - Bilateral tubal occlusion^b
 - Vasectomised partner^{b,c}

- Sexual abstinence^d
 - a Hormonal contraception may be susceptible to interaction with the study drug, which may reduce the efficacy of the contraception method.
 - b This method of contraception should preferably be used, in particular when contraception is introduced as a result of participation in the clinical study.
 - c Vasectomised partner is a highly effective birth control method provided that partner is the sole sexual partner of the WOCBP study participant and that the vasectomised partner has received medical assessment of the surgical success.
 - d Sexual abstinence is considered a highly effective method only if defined as refraining from heterosexual intercourse during the entire period of risk associated with the study treatments. The reliability of sexual abstinence needs to be evaluated in relation to the duration of the clinical study and the preferred and usual lifestyle of the patient.

APPENDIX 3: MEDICATIONS OR FOODS KNOWN TO INHIBIT OR INDUCE CYP3A

The following list provides medications known to induce or inhibit Cytochrome P450 (CYP) 3A activity. Note that this is not a comprehensive list of all medications which may modulate CYP3A activity.

NOTE: Patients receiving duvelisib are prohibited from concomitant use of medications or foods that are known to be strong inhibitors or inducers of CYP3A.

Classification of In Vivo Inhibitors of CYP3A

Strong Inhibitors ^a	Moderate inhibitors ^b	Weak inhibitors ^c
Boceprevir, clarithromycin, conivaptan, grapefruit juice ^e , indinavir, itraconazole, ketoconazole, lopinavir/ritonavir, mibefradil ^f , nefazodone, nelfinavir, posaconazole, ritonavir, saquinavir, telaprevir, telithromycin, voriconazole	Amprenavir, aprepitant, atazanavir, ciprofloxacin, darunavir/ritonavir, diltiazem, erythromycin, fluconazole, fosamprenavir, grapefruit juice ^e , imatinib, verapamil	Alprazolam, amiodarone, amlodipine, atorvastatin, bicalutamide, cilostazol, cimetidine, cyclosporine, fluoxetine, fluvoxamine, ginkgo ^d , goldenseal ^d , isoniazid, nilotinib, oral contraceptives, ranitidine, ranolazine, tipranavir/ritonavir, zileuton

a A strong inhibitor for a specific CYP is defined as an inhibitor that increases the area under the curve (AUC) of a substrate for that CYP by equal or more than 5-fold or > 80% decrease in CL.

b A moderate inhibitor for a specific CYP is defined as an inhibitor that increases the AUC of a sensitive substrate for that CYP by less than 5-fold but equal to or more than 2-fold or 50-80% decrease in CL.

c A weak inhibitor for a specific CYP is defined as an inhibitor that increases the AUC of a sensitive substrate for that CYP by less than 2-fold but equal to or more than 5-fold or 20-50% decrease in CL.

d Herbal product.

e The effect of grapefruit juice varies widely among brands and is concentration, dose, and preparation dependent. Studies have shown that it can be classified as a “strong CYP3A inhibitor” when a certain preparation was used (e.g., high dose, double strength) or as a “moderate CYP3A inhibitor” when another preparation was used (e.g., low dose, single strength).

f Withdrawn because of safety reasons.

Classification of In Vivo Inducers of CYP3A

Strong Inducers ≥ 80% Decrease in AUC	Moderate Inducers 50-80% Decrease in AUC	Weak Inducers 20-50% Decrease in AUC
Avasimibe ^a , carbamazepine, phenytoin, rifampin, St. John's Wort ^b	Bosentan, efavirenz, etravirine, modafinil, nafcillin	Amprenavir, aprepitant, armodafinil, echinacea ^c , pioglitazone, prednisone, rufinamide

a Not a marketed drug.

b The effect of St. John's Wort varies widely and is preparation dependent.

c Herbal product.

APPENDIX 4: CYP3A OR CYP2C8 SUBSTRATES

The following lists provide known sensitive CYP3A substrates, CYP3A substrates with a narrow therapeutic range, and CYP2C8 substrates. Drugs or foods that are substrates of CYP3A should only be used if medically necessary and therapeutic alternatives do not exist. Medications that are metabolised via CYP2C8 may be used as medically indicated but with caution.

Sensitive CYP3A Substrates	
Budesonide Buspirone Eplerenone Eletriptan Felodipine Fluticasone Lovastatin	Midazolam Saquinavir Sildenafil Simvastatin Triazolam Vardenafil
CYP3A Substrates with a Narrow Therapeutic Range	
Alfentanil Astemizole Cisapride Cyclosporine Diergotamine Ergotamine	Fentanyl Pimozide Quinidine Sirolimus Tacrolimus Terfenadine
CYP2C8 Substrates	
Paclitaxel Torsemide Amodiaquine	Cerivastatin Repaglinide Rosiglitazone Pioglitazone

APPENDIX 5: P-GP SUBSTRATES AND MEDICATIONS THAT ARE INHIBITORS OF P-GP

The following list provides medications that are substrates or inhibitors of P-gp, which should be used with caution during treatment with duvelisib. Note that this is not a comprehensive list of all medications which may be substrates of P-gp or may modulate P-gp activity.

P-gp Substrates	
Amitriptyline	Loperamide
Amiodarone	Losartan
Atorvastatin	Lovastatin
Cefoperazone	Methadone
Chlorpromazine	Methotrexate
Cimetidine	Methylprednisolone
Ciprofloxacin	Morphine
Clarithromycin	Nadolol
Colchicine	Norfloxacin
Cyclosporine	Nortriptyline
Dexamethasone	Ondansetron
Digoxin	Omeprazole
Diltiazem	Pantoprazole
Erythromycin	Phenytoin
Estradiol	Pravastatin
Fentanyl	Propranolol
Fexofenadine	Quinidine
Hydrocortisone	Ranitidine
Itraconazole	Sirolimus
Lansoprazole	Tacrolimus
Levofloxacin	Timolol
Lidocaine	Trimethoprim
	Verapamil
P-gp Inhibitors	
Amiodarone	Lovastatin
Amitriptyline	Mefloquine
Carvedilol	Nicardipine
Chlorpromazine	Nifedipine
Clarithromycin	Ofloxacin

Cortisol	Omeprazole
Cyclosporine	Pantoprazole
Desipramine	Progesterone
Diltiazem	Propafenone
Dipyridamole	Propranolol
Doxepin	Quinidine
Erythromycin	Rifampicin (Rifampin)
Felodipine	Saquinavir
Fluphenazine	Simvastatin
Grapefruit juice	Sirolimus
Haloperidol	Tacrolimus
Itraconazole	Testosterone

Source: [Atkinson 2007](#)

Reference:

Atkinson AJ et al. Principles of Clinical Pharmacology, 2nd ed. Academic Press, Massachusetts, 2007

**APPENDIX 6: EASTERN COOPERATIVE ONCOLOGY GROUP (ECOG) SCALE
FOR PERFORMANCE STATUS**

Grade	Description
0	Normal activity. Fully active, able to carry on all pre-disease performance without restriction.
1	Symptoms but ambulatory. Restricted in physically strenuous activity, but ambulatory and able to carry out work of a light or sedentary nature (e.g., light housework, office work).
2	In bed < 50% of the time. Ambulatory and capable of all self-care, but unable to carry out any work activities. Up and about more than 50% of waking hours.
3	In bed > 50% of the time. Capable of only limited self-care, confined to bed or chair more than 50% of waking hours.
4	100% bedridden. Completely disabled. Cannot carry on any self-care. Totally confined to bed or chair.
5	Dead.
Source: Oken 1982	

Reference:

Oken MM, Creech RH, Tormey DC, et al. Toxicity and response criteria of the Eastern Cooperative Oncology Group. Am J Clin Oncol. 1982;5(6):649-55.

APPENDIX 7: EORTC QLQ-C30

ENGLISH



EORTC QLQ-C30 (version 3)

We are interested in some things about you and your health. Please answer all of the questions yourself by circling the number that best applies to you. There are no "right" or "wrong" answers. The information that you provide will remain strictly confidential.

Please fill in your initials:

Your birthdate (Day, Month, Year):

Today's date (Day, Month, Year):

31				

	Not at All	A Little	Quite a Bit	Very Much
1. Do you have any trouble doing strenuous activities, like carrying a heavy shopping bag or a suitcase?	1	2	3	4
2. Do you have any trouble taking a <u>long</u> walk?	1	2	3	4
3. Do you have any trouble taking a <u>short</u> walk outside of the house?	1	2	3	4
4. Do you need to stay in bed or a chair during the day?	1	2	3	4
5. Do you need help with eating, dressing, washing yourself or using the toilet?	1	2	3	4

During the past week:

	Not at All	A Little	Quite a Bit	Very Much
6. Were you limited in doing either your work or other daily activities?	1	2	3	4
7. Were you limited in pursuing your hobbies or other leisure time activities?	1	2	3	4
8. Were you short of breath?	1	2	3	4
9. Have you had pain?	1	2	3	4
10. Did you need to rest?	1	2	3	4
11. Have you had trouble sleeping?	1	2	3	4
12. Have you felt weak?	1	2	3	4
13. Have you lacked appetite?	1	2	3	4
14. Have you felt nauseated?	1	2	3	4
15. Have you vomited?	1	2	3	4
16. Have you been constipated?	1	2	3	4

Please go on to the next page

During the past week:

For the following questions please circle the number between 1 and 7 that best applies to you

1 2 3 4 5 6 7

Excellent

1	2	3	4	5	6	7
---	---	---	---	---	---	---

Excellent

APPENDIX 8: EQ-5D-5L



Health Questionnaire

English version for the UK

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Under each heading, please tick the ONE box that best describes your health TODAY.

MOBILITY

- I have no problems in walking about ☐
- I have slight problems in walking about ☐
- I have moderate problems in walking about ☐
- I have severe problems in walking about ☐
- I am unable to walk about ☐

SELF-CARE

- I have no problems washing or dressing myself ☐
- I have slight problems washing or dressing myself ☐
- I have moderate problems washing or dressing myself ☐
- I have severe problems washing or dressing myself ☐
- I am unable to wash or dress myself ☐

USUAL ACTIVITIES (e.g. work, study, housework, family or leisure activities)

- I have no problems doing my usual activities ☐
- I have slight problems doing my usual activities ☐
- I have moderate problems doing my usual activities ☐
- I have severe problems doing my usual activities ☐
- I am unable to do my usual activities ☐

PAIN / DISCOMFORT

- I have no pain or discomfort ☐
- I have slight pain or discomfort ☐
- I have moderate pain or discomfort ☐
- I have severe pain or discomfort ☐
- I have extreme pain or discomfort ☐

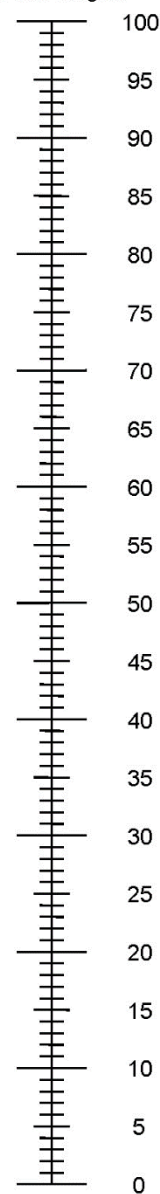
ANXIETY / DEPRESSION

- I am not anxious or depressed ☐
- I am slightly anxious or depressed ☐
- I am moderately anxious or depressed ☐
- I am severely anxious or depressed ☐
- I am extremely anxious or depressed ☐

- We would like to know how good or bad your health is TODAY.
- This scale is numbered from 0 to 100.
- 100 means the best health you can imagine.
0 means the worst health you can imagine.
- Mark an X on the scale to indicate how your health is TODAY.
- Now, please write the number you marked on the scale in the box below.

YOUR HEALTH TODAY =

The best health
you can imagine



The worst health
you can imagine

APPENDIX 9: EORTC QLQ-NHL-HG29

ENGLISH



EORTC QLQ-NHL-HG29

Patients sometimes report that they have the following symptoms or problems. Please indicate the extent to which you have experienced these symptoms or problems during the past week.

Please answer by circling the number that best applies to you.

During the past week:	Not at All	A Little	Quite a Bit	Very Much
31. Have you had muscle weakness?	1	2	3	4
32. Have you had aches or pains in your muscles or joints?	1	2	3	4
33. Have you had aches or pains in your bones?	1	2	3	4
34. Have you had a dry cough?	1	2	3	4
35. Have you had a dry mouth?	1	2	3	4
36. Have you had problems with your sense of taste?	1	2	3	4
37. Have you felt ill or unwell?	1	2	3	4
38. Have you had tingling hands or feet?	1	2	3	4
39. Have you had numbness in your fingers or toes?	1	2	3	4

During the past week:	Not at All	A Little	Quite a Bit	Very Much
40. Have you had shortness of breath on exertion?	1	2	3	4
41. Have you felt you had setbacks in your physical condition?	1	2	3	4
42. Have you had a lack of energy?	1	2	3	4
43. Have you felt drowsy?	1	2	3	4
44. Have you had sudden tiredness?	1	2	3	4

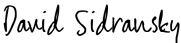
During the past week:	Not at All	A Little	Quite a Bit	Very Much
45. Have you had mood changes?	1	2	3	4
46. Have you felt a lack of confidence in your body?	1	2	3	4
47. Have you been dissatisfied with how your body functions?	1	2	3	4
48. Have you had difficulty accepting limitations due to the disease?	1	2	3	4

Please go on to next page

ENGLISH

During the past <u>4 weeks</u> :	Not at All	A Little	Quite a Bit	Very Much
49. Have you worried about picking up an infection?	1	2	3	4
50. Have you worried about your health in the future?	1	2	3	4
51. Have you worried about recurrence of your disease?	1	2	3	4
52. Have you worried about becoming chronically ill?	1	2	3	4
53. Have you worried about becoming dependent on others?	1	2	3	4
54. Have you worried about getting another type of cancer?	1	2	3	4
55. Have you worried about your treatment causing future health problems?	1	2	3	4
56. Have you worried about damage to your heart and blood vessels?	1	2	3	4
57. <u>If applicable</u> : Have you had problems at your work or place of study due to the disease?	1	2	3	4
58. <u>If applicable</u> : Have you worried about not being able to continue working or your education?	1	2	3	4
59. <u>If applicable</u> : Have you been concerned about your ability to have children?	1	2	3	4

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